

Cross-modal interaction of Alpha Activity does not reflect inhibition of early sensory processing: A frequency tagging study using EEG and MEG

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This manuscript addresses the role of alpha oscillations in sensory gain control. The authors use an attention-cuing task in an initial EEG study followed by a separate MEG replication study to demonstrate that whilst (occipital) alpha oscillations are increased when anticipating an auditory target, so is visual responsiveness as assessed with frequency tagging. The authors propose that their results demonstrate a general vigilance effect on sensory processing and offer a re-interpretation of the inhibitory role of the alpha rhythm. While some concerns remain about the interpretation of the alpha inhibition hypothesis, these results are **valuable**, and the provided evidence is **solid**.

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Abstract

Selective attention involves prioritizing relevant sensory input while suppressing irrelevant stimuli. It has been proposed that oscillatory alpha-band activity (~10 Hz) aids this process by functionally inhibiting early sensory regions. However, recent studies have challenged this notion. Our EEG and MEG studies aimed to investigate whether alpha oscillations serve as a ‘gatekeeper’ for downstream signal transmission. We first observed these effects in an EEG study and then replicated them using MEG, which allowed us to localize the sources.

We employed a cross-modal paradigm where visual cues indicated whether upcoming targets required visual or auditory discrimination. To assess inhibition, we utilized frequency-tagging, simultaneously flickering the fixation cross at 36 Hz and playing amplitude-modulated white noise at 40 Hz during the cue-to-target interval.

Consistent with prior research, we observed an increase in posterior alpha activity following cues signalling auditory targets. However, remarkably, both visual and auditory frequency

tagged responses amplified in anticipation of auditory targets, correlating with alpha activity amplitude. Our findings suggest that when attention shifts to auditory processing, the visual stream remains responsive and is not hindered by occipital alpha activity. This implies that alpha modulation does not solely regulate ‘gain control’ in early visual areas but rather orchestrates signal transmission to later stages of the processing stream.

Introduction

In our daily life, we are often confronted with sensory information from many different sources, all at once. To operate effectively, we require selective attention, reconciling the tension between environmental inputs relevant for top-down goals and sensory information that may be perceptually salient but task-irrelevant. Previous research gave rise to the prominent alpha inhibition hypothesis, which suggests that oscillatory activity in the alpha range (~10 Hz) plays a mechanistic role in selective attention through functional inhibition of irrelevant cortices (see **Fig. 1A**; [Foxe et al., 1998](#); [Jensen & Mazaheri, 2010](#); [Klimesch et al., 2007](#)). Functional inhibition refers to an area of the cortex being actively hindered to process input, which is distinctly different from idling, where a part of the cortex is not actively involved. Evidence supporting this theory revealed an increase in alpha-power over task-irrelevant sensory cortices after the onset of cues indicating the spatial location ([Kelly et al., 2006](#); [Okazaki et al., 2014](#); [Thut et al., 2006](#); [Worden et al., 2000](#); [Zumer et al., 2014](#)) or specific modality of an upcoming target ([Foxe et al., 1998](#); [Fu et al., 2001](#); [Mazaheri et al., 2014](#)). Moreover, previous investigations have observed ‘spontaneous fluctuations’ of alpha power in sensory regions, particularly in the visual cortex, to be inversely related to discrimination ability ([Ergenoglu et al., 2004](#); [Van Dijk et al., 2008](#)). Alpha inhibition is believed to be transmitted in a phasic manner, as phosphene perception as well as high-frequency and spiking activity vary in line with the alpha cycle ([Dugué et al., 2011](#); [Haegens et al., 2011](#); [Spaak et al., 2012](#)).

Recent evidence challenged a direct connection between alpha activity and visual information processing in early visual cortex. As such, both visual steady-state responses and alpha power were modulated by attention, but did not covary when investigating individual trials ([Zhigalov & Jensen, 2020](#)). Unfortunately, very few studies have investigated direct connections between alpha activity, attention and sensory signals, especially over trials. Furthermore, results seem to depend on timing of alpha activity in relation to sensory responses as well as stimulus type and outcome measure ([Morrow et al., 2023](#)).

Accordingly, the objective of the current study is to test the alpha inhibition hypothesis compared to an alternative theory. Based on the alpha inhibition hypothesis, alpha modulation is connected to ‘gain control’ in early visual areas through modulation of excitability ([Foxe & Snyder, 2011](#); [Jensen & Mazaheri, 2010](#); [Van Diepen et al., 2019](#)). In contrast, we propose that inhibitory effects of alpha modulation are exhibited at later stages of the processing stream ([Peylo et al., 2021](#); [Yang et al., 2023](#); [Zhigalov & Jensen, 2020](#); [Zumer et al., 2014](#)), gating feedforward or feedback communication between sensory areas (see **Fig. 1**; [Bauer et al., 2020](#); [Haegens et al., 2015](#); [Uemura et al., 2021](#)).

To this end, we applied frequency-tagging, the rhythmic presentation of sensory stimuli, which elicits steady-state sensory evoked potentials or fields (SSEP/SSEF), consisting of rhythmic neuronal activity in the frequency of stimulation ([Brickwedde et al., 2020](#); [Colon et al., 2012](#); [Dinse et al., 1998](#); [Marzoll et al., 2018](#); [Regan, 1982](#); [Snyder, 1992](#); [Stapells et al., 1984](#); [Tobimatsu et al., 1999](#)). The magnitude of SSEPs is attention-dependent ([de Jong et al., 2010](#); [Müller et al., 1998](#); [Müller & Hillyard, 2000](#); [Porcu et al., 2013](#); [Saupe et al., 2009](#); [Toffanin et al., 2009](#)) even for frequencies too fast to perceive consciously ([Brickwedde et al., 2022](#); [Zhigalov et al., 2019](#)). Visual SSEP/SSEFs are most strongly observable over occipital areas, whereas auditory SSEP/SSEFs appear most strongly over temporal (MEG) or fronto-to-central (EEG)

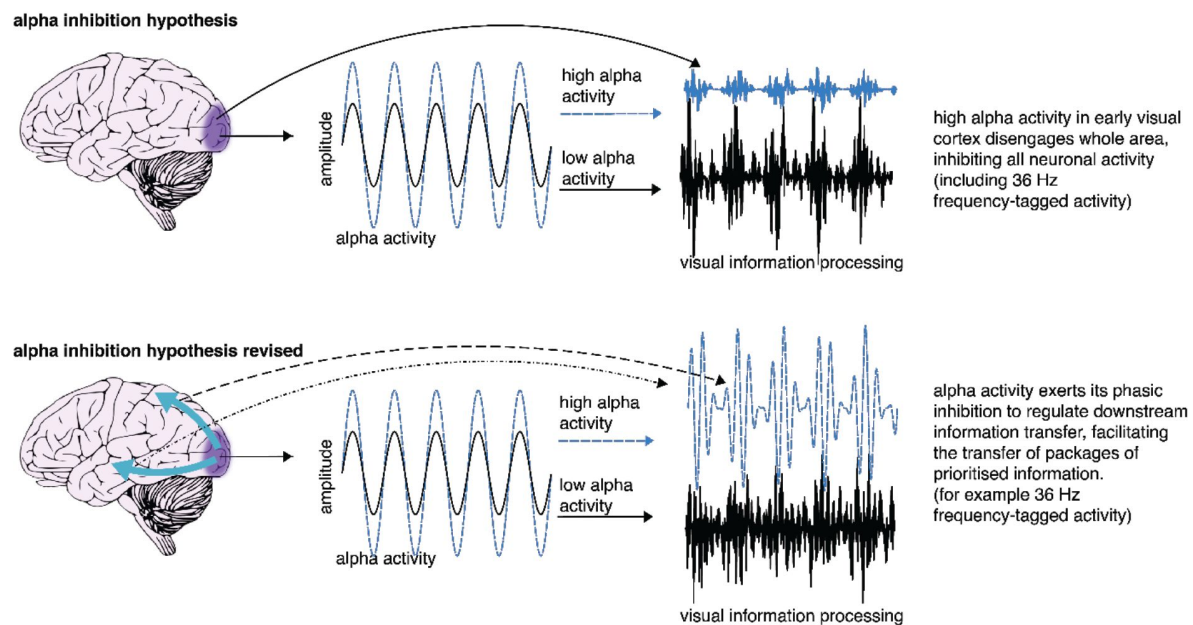


Figure 1.

Illustration of the alpha inhibition hypothesis.

The alpha inhibition hypothesis suggests that occipital alpha inhibits visual information processing in a phasic manner. If alpha activity is high, it suggests that the whole area is inhibited and thereby disengaged (Foxe et al., 1998 [\[1\]](#); Jensen & Mazaheri, 2010 [\[2\]](#); Klimesch et al., 2007 [\[3\]](#)). We propose a revision of this hypothesis, whereby alpha activity exerts its phasic inhibition to regulate downstream information transfer, creating enhanced signal packages of prioritised information (see also Yang et al., 2023 [\[4\]](#); Zhigalov & Jensen, 2020 [\[5\]](#); Zumer et al., 2014 [\[6\]](#)).

areas (de Jong et al., 2010 [↗](#); Hari et al., 1989 [↗](#); Pantev et al., 1996 [↗](#); Regan, 1982 [↗](#)). Furthermore, when applying two different frequencies for two different sensory modalities, their intermodulation frequency (f_1 - f_2) has been suggested to reflect cross-modal integration (Drijvers et al., 2021 [↗](#)). Due to distinct responses, localisation and attention-dependence, frequency-tagging provides an optimal tool to study sensory signal processing and integration over time.

The aim of our study was to directly test the alpha inhibition hypothesis by investigating if cue-induced modulation of alpha activity coincides with the suppression of frequency-tagging responses in task-irrelevant modalities. Based on previous studies, we utilized a cross-modal attention paradigm, in which symbolic visual cues signalled the target modality (visual or auditory) of an upcoming discrimination task (e.g., van Diepen et al., 2015 [↗](#)). Here, we included an additional experimental manipulation in the form of frequency-tagging to assess the involvement of the auditory and visual systems in the *cue-to-target interval*.

In line with previous results, we hypothesized that signalling an upcoming auditory target would lead to increased alpha activity over visual occipital regions as well as increased SSEP responses to auditory and decreased SSEP to visual stimuli as indexed by frequency-tagging. In brief, while we observed the expected cue-induced early-visual alpha modulation, the amplitude of auditory and visual SSEP/SSEFs as well as their intermodulation frequency increased just prior to the onset of the auditory target, contradicting the alpha inhibition hypothesis. The difference between conditions of visual SSEP/SSEFs originated from sensory integration areas and correlated with early sensory alpha activity on a trial-by-trial basis, speaking to an effect of alpha modulation on signal transmission rather than inhibition of early visual areas.

Results

Results Study 1 – cross-modal EEG experiment

To assess audio-visual excitability in anticipation of either visual or auditory targets, we displayed visual cues to signal the modality of the upcoming target (auditory, visual, or non-specific). In a three-second cue-to-target interval, we frequency-tagged the fixation cross at 36 Hz and played 40 Hz amplitude modulating white noise. Immediately afterwards, participants had to either discriminate between three different pitch sounds (auditory target) or three different Gabor patch orientations (visual target). If the target modality was cued (e.g. not non-specific), 50% of the trials were accompanied by a random distractor from the target pool of the opposing sensory modality (see Fig. 2A [↗](#)).

Behavioural performance

We found that accuracy differed significantly between conditions ($F_{(5,105)} = 44.16$; $p < .001$). Overall, participants were significantly less accurate in the auditory discrimination task ('overall auditory', $M = 79\%$ correct, $SD = 11.7$) than in the visual discrimination task ('overall visual', $M = 97\%$, $SD = 2.3$; see Fig. 2B [↗](#)), with the worst performance occurring when auditory targets were paired with visual distractors (auditory +: $M = 74\%$ correct, $SD = 13$).

Reaction times yielded a similar pattern, with auditory reactions (overall auditory: $M = 662$ ms, $SD = 136$) being slower than for the visual task (overall visual: $M = 597$ ms, $SD = 130$; main effect over all conditions: $F_{(5,105)} = 27.47$; $p < .001$; see Fig. 2C [↗](#)). This was mostly driven by slow responses to auditory targets paired with distractors (auditory +: $M = 723$ ms).

To ascertain that cues and distractors were functionally relevant, we calculated attentional benefit (cued - non-specific condition) and distractor cost (cued condition with distractor – cued condition without distractor). As expected, cues improved reaction times over all conditions and distractors

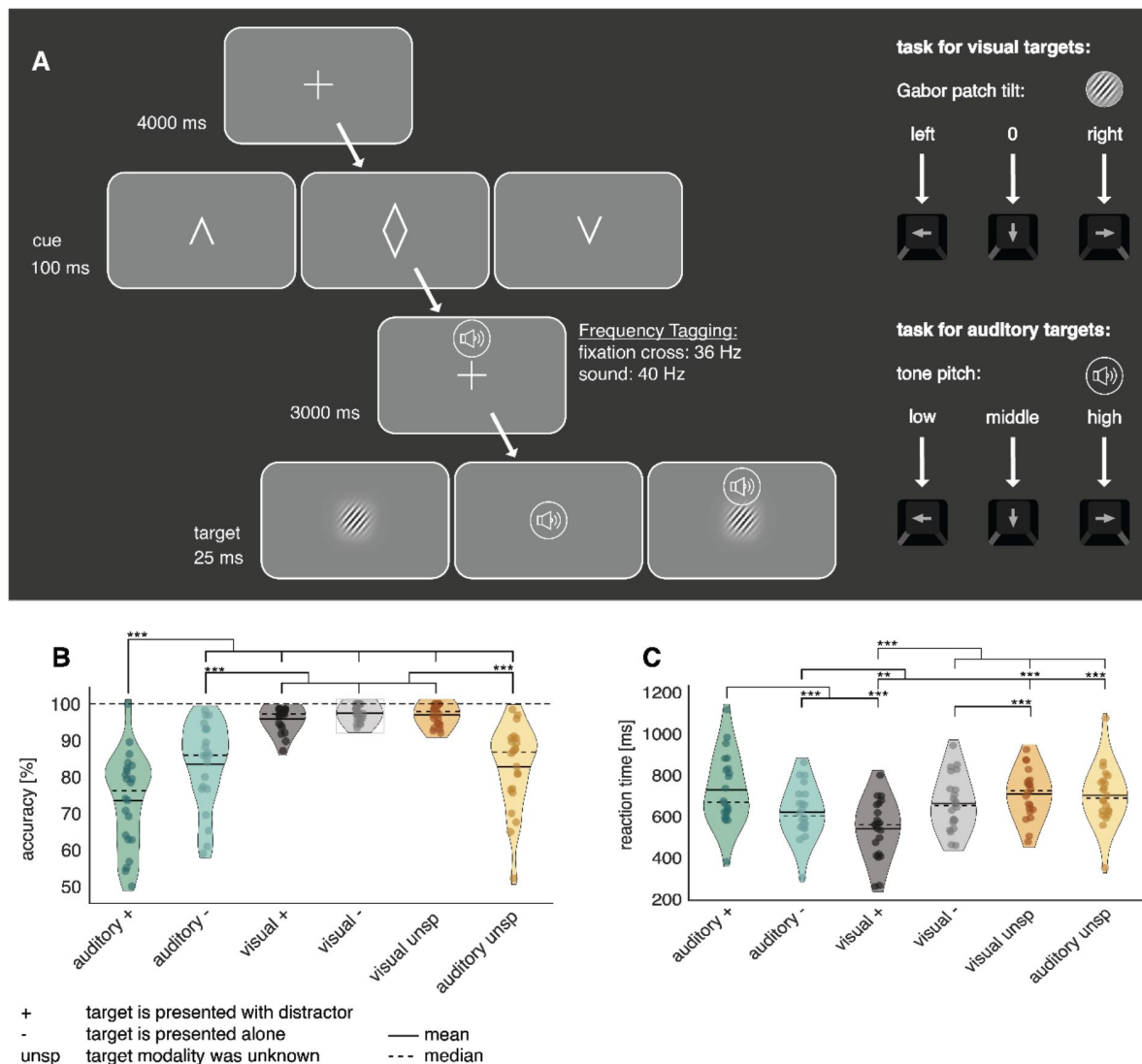


Figure 2.

A, Illustration of the cross-model discrimination task in Study 1. Trials were separated by a 4 s interval, in which a fixation cross was displayed. A brief central presentation of the cue (100 ms) initiated the trial, signalling the target modality (see figure above from left to right: auditory, non-specific, visual). In the cue-to-target interval, the fixation cross was frequency-tagged at 36 Hz. At the same time, a sound was displayed over headphones, which was frequency-tagged at 40 Hz. Both tones and fixation cross contained no task-relevant information. The target, consisting either of a static Gabor patch or a tone, was presented for 25 ms. Participants had to differentiate via button presses between 3 different Gabor rotations or tone pitches, respectively. In 50% of auditory and visually cued trials, a distractor in form of a random pitch or rotation of the un-cued modality was presented alongside the target. B-C, Analysis of task accuracy and reaction time indicates increased difficulty of auditory targets. B, Task accuracy compared between all 6 experimental conditions reveals a drop in accuracy for responses to auditory targets. C, reaction times of correct trials compared between all 6 experimental conditions. The slowest reaction times are observable following auditory targets alongside visual distractors. N = 22; *** sig. < .001; ** sig. < .01; * sig. < .05.

impaired the accuracy over all conditions and reaction times for auditory targets. (see [suppl Fig. 1](#)). Interestingly, auditory distractors reduced reaction times to visual targets, which could be explained by a generally faster processing of auditory targets (Jain et al., 2015). As such, the auditory distractor possibly caused intersensory facilitation (Nickerson., 1973), whereby reaction times to a target can be facilitated when accompanied by stimuli of other sensory modalities, even if they are irrelevant or distracting.

Cross-modal cues differentially modulated pre-target alpha activity

We conducted a time-frequency analysis of power in the cue-to-target interval and found a stronger amplitude increase from baseline for auditory compared to visual target conditions starting around 2 s before target onset ([Fig. 3](#)). We calculated the time course of alpha power changes using the Hilbert-transformation (8 – 12 Hz, [Fig. 3A-B](#)). We then applied cluster permutation analysis, whereby real condition differences were tested against coincidental findings by randomly permutating the condition labels to the data and testing for condition differences 1000 times (Maris & Oostenveld, 2007). Consistent with previous work (Mazaheri et al., 2014; van Diepen & Mazaheri, 2017), analysing the last two seconds before target onset revealed two significant clusters of difference in alpha power when expecting an auditory compared to a visual target. These effects corresponded to clusters extending from -1.84 to -0.64 s ($p = .004$) and -0.62 to 0s ($p = .005$). When presenting non-specific cues, alpha power changes were not significant, but descriptively larger compared to visual target conditions and lower compared to auditory target conditions (see [suppl Fig. 2](#)). However as significant alpha modulation was a prerequisite to test our hypotheses, we excluded this condition from further analysis.

Cross-modal cues increased the amplitude of the frequency-tagged responses across both modalities

To assess the temporal development of frequency-tagging responses, steady-state potentials were calculated using data band-pass filtered around the tagging frequency. Neuronal 40 Hz auditory-steady state evoked potential (ASSEP) responses were strongest over central areas ([Fig. 4A](#)), and 36 Hz auditory-steady state evoked potential (VSSEP) responses were strongest over occipital areas ([Fig. 4B](#)).

To assess the differences between conditions, the Hilbert envelope of the steady-state potentials was analysed using cluster permutation analyses. When expecting an auditory target compared to a visual target, the 40 Hz ASSEP response was larger shortly before target onset (see [Fig. 4C](#); -0.15 to -0.08 s; $p = .041$).

Surprisingly, the 36 Hz VSSEP response was likewise increased shortly before expecting an auditory compared to a visual target (see [Fig. 4D](#); -0.16 to -0.06 s; $p = .014$). As such, the observed frequency tagging responses might reflect effort, which affects the vigilance of the whole sensory system rather than sensory-specific allocations of attention. For both 36 Hz VSSEP and 40 Hz ASSEP responses, condition differences appeared strongest over mid-parietal regions in contrast to primary sensory cortices.

Alpha power was positively correlated with amplitude of frequency tagged responses

Following the observation of condition differences in alpha activity and frequency-tagging responses, we were further interested in exploring whether these responses were connected. Accordingly, we conducted trial-by-trial correlations using alpha condition differences and their electrode positions as seed, which was correlated with frequency-tagging signals over all electrodes. Multiple comparison correction was applied by testing the correlation matrix against a zero-correlation matrix with a cluster permutation approach. A positive correlation was observed over right parietal-to-occipital areas between the late alpha cluster activity and both 40 Hz ASSEP

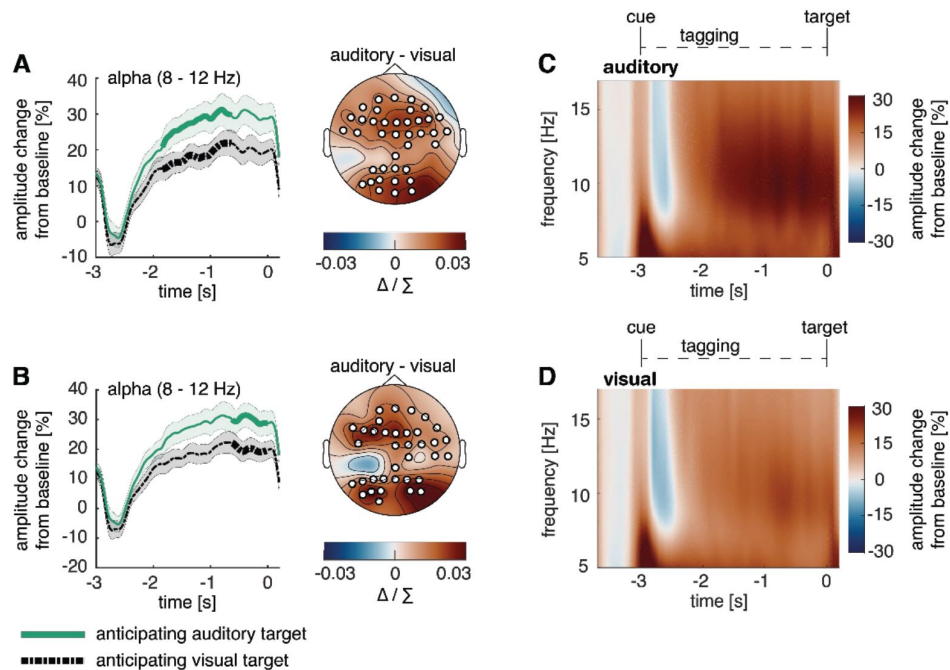


Figure 3.

Post-cue modality specific modulation of alpha power in anticipation of an auditory versus a visual target.

A-B, the time course of post-cue alpha power. Cluster permutation analysis resulted in two condition effects, both indicating heightened alpha activity when expecting an auditory compared to a visual target (A: $p < .01$; B: $p < .01$). C-D, Time-frequency representation of power in the cue-to-target interval. A greater increase in alpha power was observed when expecting an auditory target (average over significant electrodes for the condition difference in A). note: Cluster electrodes are marked in white. Shading represents standard error from the mean; Δ / Σ represents $(a-b)/(a+b)$ normalization.

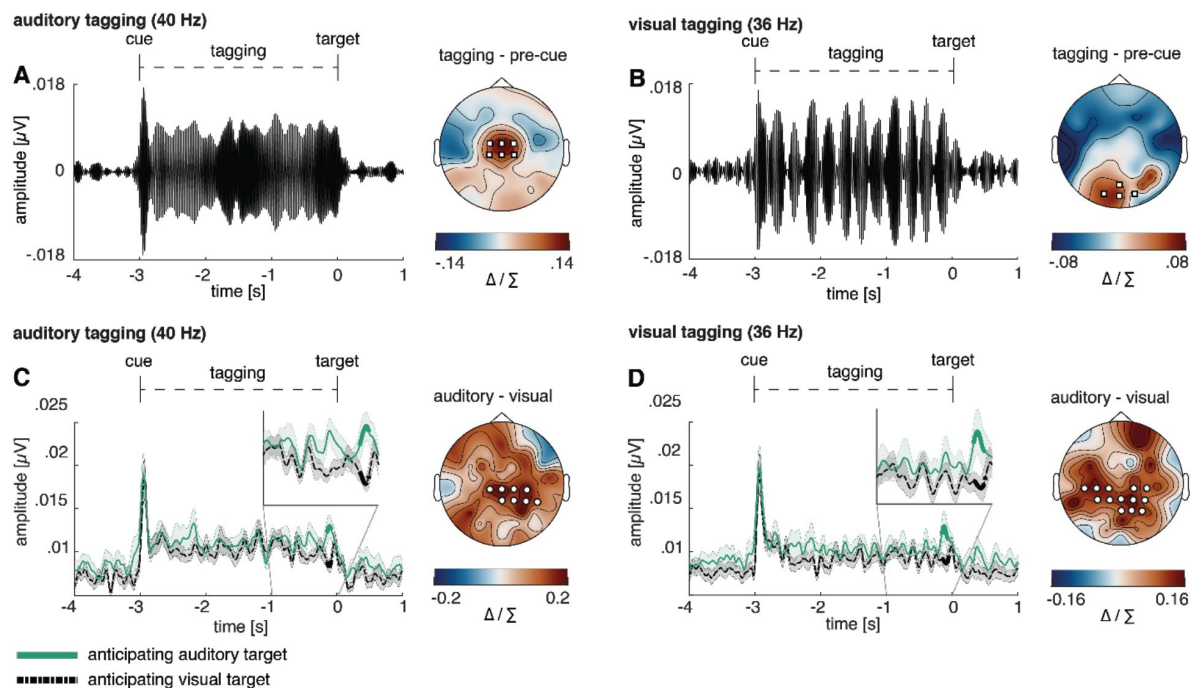


Figure 4.

Increase in amplitude of both visual and auditory frequency tagged responses when anticipating visual or auditory targets.

Event-related potentials and scalp topographies reveal distinct modality specific responses at the tagged frequencies. A, auditory steady-state evoked potential (ASSEP) averaged over 6 central electrodes displaying the highest 40 Hz power (Cz, FC1, FC2, C1, C2, FCz; marked with white squares). B, visual steady-state evoked potential (VSSEP) averaged over 4 occipital electrodes displaying the highest 36 Hz power (POz, O1, O2, Oz; marked with white squares) C, the Hilbert-envelope of the 40 Hz ASSEP reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p = .041$); D, the Hilbert-envelope of the 36 Hz VSSEP likewise reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p = .014$). note: Cluster electrodes are marked in white. Shading represents standard error from the mean. Δ / Σ represents $(a-b)/(a+b)$ normalization.

($p = .009$) and 36 Hz VSSEP ($p = .004$) responses when expecting a visual target (see **Fig. 5A-B**). To estimate the robustness of these results, we additionally conducted median split analyses between trials with high and low alpha power for each participant, as well as averaged the correlation coefficient of each participant and calculated a one-sample t-test against 0. For each analysis we provided the Bayes Factor, which estimates the strength of support for or against the null hypothesis ($BF > 3.2$ is considered as substantial evidence and $BF > 10$ is considered as strong evidence; Kass & Raftery, 1995).

The median split was highly significant for the 36 Hz VSSEP response (**Fig. 5A**, middle columns, $p = .033$; $t_{(19)} = 2.29$; $BF_{(10)} = 1.91$) but did not reach significance for the 40 Hz ASSEP response (**Fig. 5B**, middle column; $p = 0.20$; $t_{(19)} = 1.32$; $BF_{(10)} = 0.49$). Correlation coefficients indicated strong correlations with alpha activity for both 40 Hz ASSEP (**Fig. 5B**, right column; $p < .001$; $t_{(19)} = 4.95$; $BF_{(10)} = 306.93$) and 36 Hz VSSEP activity (**Fig. 5A**, right column; $p < .01$; $t_{(19)} = 3.66$; $BF_{(10)} = 23.57$).

The same positive correlation with alpha activity was found when expecting an auditory target for 36 Hz VSSEP activity ($p = .031$), but not for 40 Hz ASSEP activity (see **Fig. 5D**). A median split between high and low alpha activity ($p = .005$; $t_{(19)} = 3.14$; $BF_{(10)} = 8.53$) and correlation coefficients ($p = .002$; $t_{(19)} = 3.52$; $BF_{(10)} = 17.76$) provided moderate to strong evidence for this effect. The likely origin of alpha modulations from early visual cortices may explain a stronger connection to 36 Hz VSSEP responses compared to 40 Hz ASSEP responses.

Additionally, we compared how correlation coefficients between alpha activity and frequency-tagging differed when anticipating an auditory versus a visual target. Multiple comparison correction was applied with cluster permutation analysis. Interestingly, an interaction between the strength of the correlation associating alpha, 36 Hz VSSEP activity and condition became apparent ($p = .044$; see **Fig. 5E**) and was observed most strongly over right-central electrodes. Comparing the correlation coefficients of participants over this cluster revealed a strong effect and even a change of direction in the correlation ($p < .001$; $t_{(21)} = -4.76$; $BF_{(10)} = 259.97$). Particularly, when expecting a visual target, there was a negative correlation between 36 Hz VSSEP and alpha activity, which turned positive when expecting an auditory target. Both correlations also differed significantly from 0 (expecting a visual target: $p < .001$; $t_{(21)} = -3.87$; $BF_{(10)} = 39.62$; expecting an auditory target: $p = .020$; $t_{(21)} = 2.53$; $BF_{(10)} = 2.83$). In contrast to the previously observed positive correlation between alpha activity and 36 Hz VSSEP activity, the significant electrode cluster was located more ventrally. This effect could possibly hint at dynamic adaptability of oscillatory alpha effects on later processing stages.

It is further noteworthy that the correlation between alpha activity and 36 Hz VSSEP response when expecting a visual target, was also present when utilizing the early alpha cluster as seed (~1200 to 400 ms before target onset, see **Fig. 5C**), in which case alpha modulation greatly preceded the 36 Hz VSSEP modulation ($p = .016$). For this correlation, the median split between high and low alpha trials did not reach significance ($p = .11$; $t_{(20)} = 1.69$; $BF_{(10)} = 0.76$), however testing the correlation coefficients against 0 again revealed a significant effect ($p = .003$; $t_{(20)} = 3.39$; $BF_{(10)} = 14.22$). This result may support the directionality of alpha modulation affecting visual information processing (and not vice versa).

Intermodulation frequency

Lastly, we analysed the steady-state response of the intermodulation frequency at 4 Hz. Increased intermodulation shortly before target onset could be observed when expecting an auditory compared to a visual target (-0.51 to -0.0620 s; $p < .001$). This effect was strongest over left fronto-to-central electrodes and right central-to-occipital electrodes (see **Fig. 6A**) and may suggest in increase in cross-modal integration.

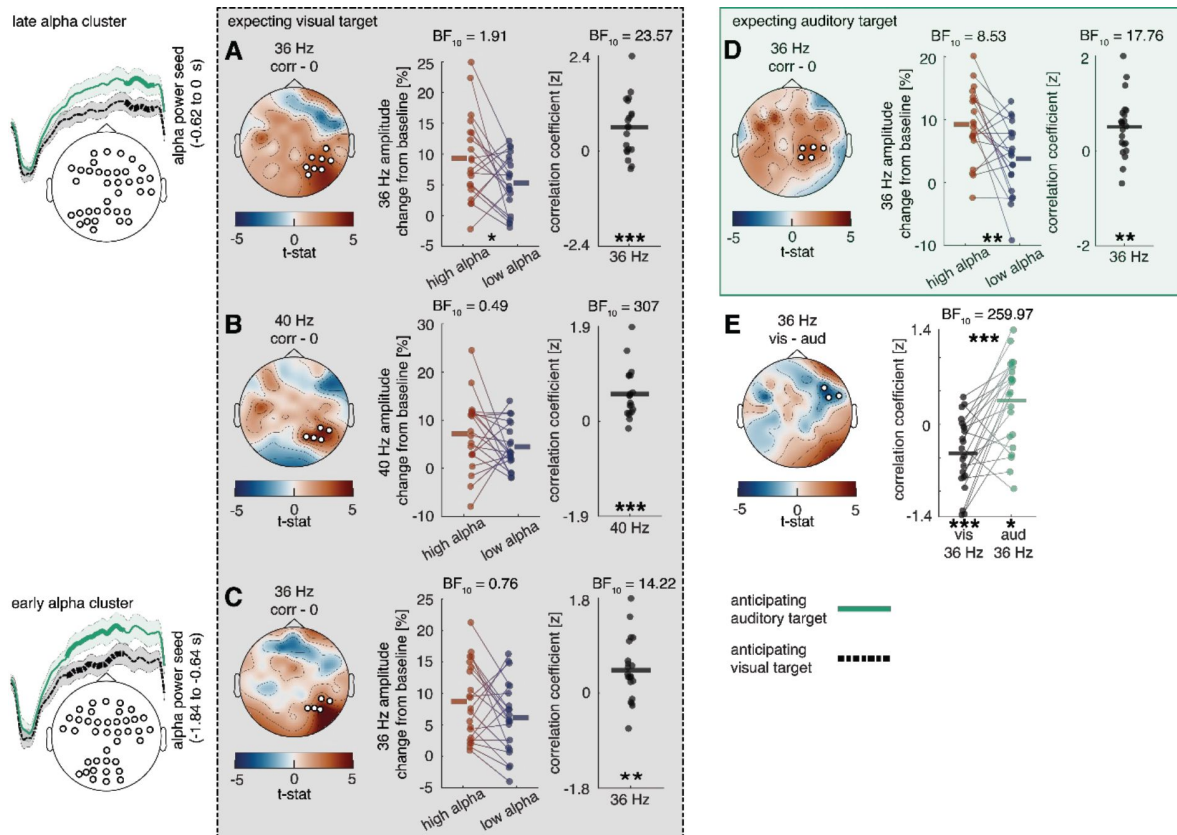


Figure 5.

Relationship between cue induced alpha modulation and amplitude of frequency tagged responses in study 1.

Previously obtained alpha clusters (see Fig. 3) were correlated over trials with auditory 40 Hz and visual 36 Hz clusters (see Fig. 4), where alpha electrodes/sensors were applied as seeds. The analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically. Additionally, median splits between high and low alpha trials as well as correlation coefficients of these clusters are displayed for all participants A-B, a positive correlation is visible between alpha activity in the last 400 ms and steady state potentials shortly before target onset when expecting a visual target (visual 36 Hz: $p = .013$; 40 Hz: $p = .009$). D, when expecting an auditory target, there is a positive correlation between alpha activity in the last 400 ms and visual 36 Hz activity shortly before target onset ($p = .010$). E, the correlation between alpha activity 400 ms and visual 36 Hz activity shortly before target onset changes its direction depending on whether an auditory or a visual target is expected ($p = .037$). C, a positive correlation is also visible between alpha activity as early as ~1200 ms to 400 ms and visual 36 Hz activity shortly before target onset when expecting a visual target ($p = .016$). $N = 22$; *** sig < .001; ** sig < .01; * sig < .05. + sig. < .1.

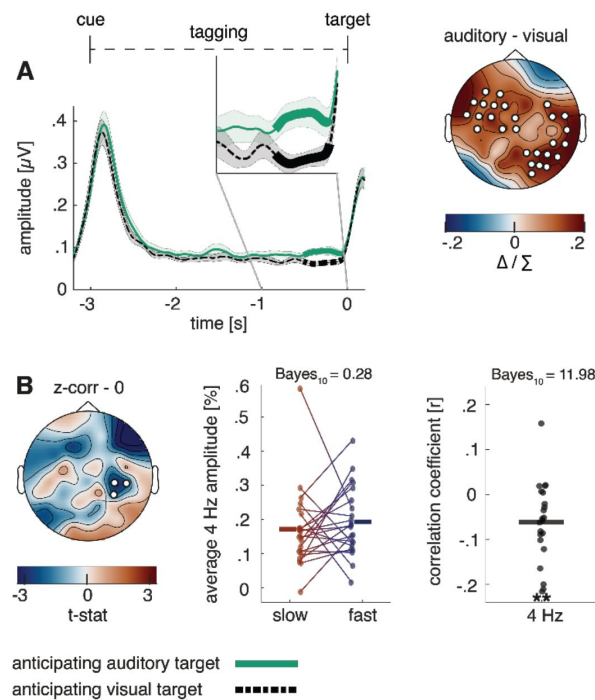


Figure 6.

Steady-state response in the intermodulation frequency and its behavioural relevance.

A, the Hilbert-envelope of the 4 Hz steady-state response reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p < .01$). B, there is a trial-by-trial correlation between 4 Hz activity and reaction time when a visual target without distractor was presented. The correlation is further illustrated by a median split between fast and slow reaction time trials as well as by correlation coefficients for each participant. $N = 22$; ** sig. $< .01$.

With the goal to examine whether there are any behavioral effects of the condition difference, trial by trial correlations with reaction times of each of the 2 auditory and visual conditions were performed. Only in the easiest condition, when expecting a visual target that was not accompanied by a distractor, a negative correlation with reaction time could be found, strongest over right central electrodes (see **Fig. 6B**; $p = .046$). While a median split between slow and fast trials did not reach significance ($p = .50$; $t_{(21)} = -0.69$; $BF_{(10)} = .28$), testing the correlation coefficients against 0 revealed strong evidence for a correlation ($p = .004$; $t_{(21)} = -3.28$; $BF_{(10)} = 11.98$). If the intermodulation frequency reflects cross-modal integration (Drijvers et al., 2021), this effect would indicate faster reaction times for trials with stronger sensory integration.

Results Study 2 – cross-modal MEG experiment

To test the robustness of our results and to employ additional control analyses, we replicated our experiment using MEG (see **Fig. 7A**). While an increase in visual information processing parallel to an increase in alpha modulation already contradicts the notion of alpha inhibition exerting “gain control”, affecting the whole visual cortex, our claim that alpha modulation instead affects visual information at later processing stages still required further validation. As such, our goal was to perform source analyses showing alpha modulation originating from primary visual areas affected visual information at later processing stages (e.g. not in primary visual cortex). Additionally, to exclude that the uncertainty over possible distractors affected our results, we employed a block design, where block 1 consisted only of trials without distractors and in block 2 targets were always accompanied by a distractor. Furthermore, we aligned the visual and auditory task to be more similar, both of them now featuring frequency-discrimination, which related to sound pitch (frequency) in the auditory condition and stripe-frequency of the Gabor patch in the visual condition. Lastly, to make sure our effects were driven by sensory modality-differences rather than task-difficulty differences, we included a short calibration phase. Prior to the experiment, difficulty of pitch sounds, and Gabor patch frequency were calibrated for each individual, ascertaining a success rate between 55% to 75%.

Behavioural performance

Our adjustments in the MEG-study streamlined performances to be more in line between auditory and visual conditions in the second block, however in the first block there were strong condition differences (Main effect condition: $F_{(3,75)} = 10.26$; $p < .01$). In block 1, participants were significantly more accurate in the auditory (‘block 1 auditory’, $M = 84\%$ correct, $SD = 10.03$) compared to the visual task (‘block 1 visual’, $M = 70\%$, $SD = 10.61$; see **Fig. 7B**). However, in block 2, there was no observable difference between visual and auditory task accuracy (‘block 2 auditory’, $M = 77\%$ correct, $SD = 10.03$; ‘block 2 visual’, $M = 78\%$, $SD = 13.15$). Reaction times were comparable between conditions (overall: $M = 661$ ms, $SD = 76$), with the exception of responses to visual targets in block 1 (block 1 visual: $M = 726$ ms, $SD = 85$), which were significantly slower than all other reaction times (main effect condition: $F_{(3,75)} = 18.13$; $p < .001$; see **Fig. 7C**).

Due to the block design, distractor cost could not be differentiated from learning effects. For auditory targets, there was a significant decrease in accuracy between block 1 and 2, while for visual targets, there was a significant increase. Similar to study 1, reaction times for auditory targets stayed roughly the same, but became shorter for visual targets (see **suppl Fig. 1**).

Cross-modal cues differentially modulated pre-target alpha activity in early visual areas

In line with previous studies (van Diepen & Mazaheri, 2017), condition differences in alpha activity were only significant in block 2, where distractors were always present. As alpha modulation was a prerequisite to test our hypotheses, we performed the following analyses solely with data from block 2 (see **Fig. 8**).

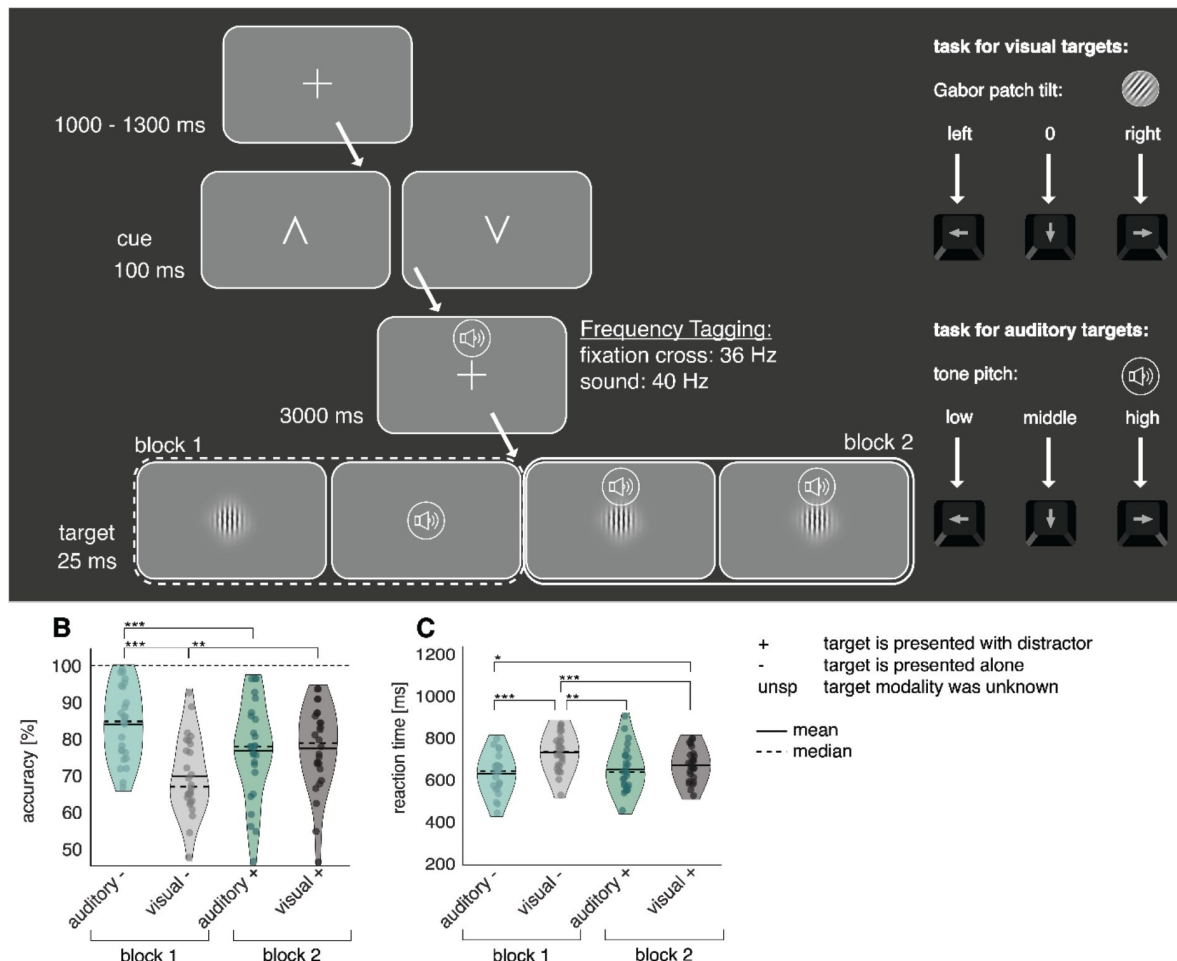


Figure 7.

A, Illustration of the cross-model discrimination task in Study 2. Trials were separated by a 1-1.3 s interval, in which a fixation cross was displayed. A brief central presentation of the cue (100 ms) initiated the trial, signaling the target modality (from left to right: auditory, visual). In the cue-to-target interval, the fixation cross was frequency-tagged at 36 Hz. At the same time, a sound was displayed over headphones, which was frequency-tagged at 40 Hz. Both tones and fixation cross contained no task-relevant information. The target, consisting either of a static Gabor patch or a tone, was presented for 25 ms.

Participants had to differentiate via button presses between 3 different Gabor stripe frequencies or tone pitches, respectively. In block 2, a distractor in form of a random sound or Gabor patch of the un-cued modality was always presented alongside the target. B-C, analysis of task accuracy and reaction time indicates comparable difficulties in block 2. B, task accuracy differences were only observable in the first block without distractors. C, reaction times to visual targets in the first block were strongly increased compared to all other conditions. In the second block, no significant difference in reaction times was observable. N = 27; *** sig. < .001; ** sig. < .01; * sig. < .05.

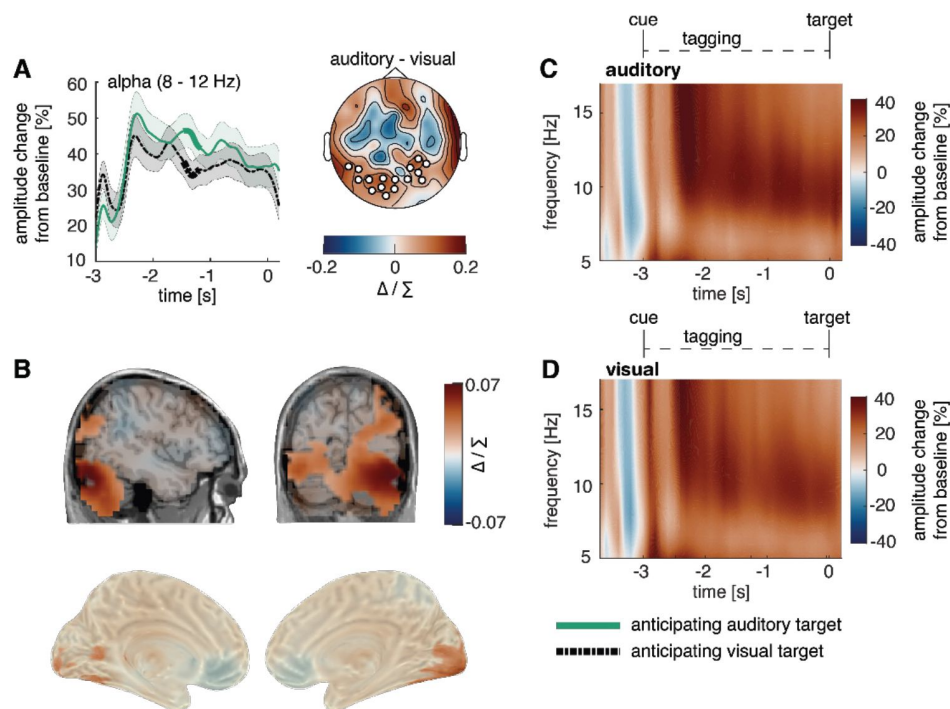


Figure 8.

Post-cue modality specific early visual modulation of alpha power in anticipation of an auditory versus a visual target.

A, the time course of post-cue alpha power. Cluster permutation analysis resulted in a condition effects, indicating heightened alpha activity when expecting an auditory compared to a visual target ($p = .034$). B, source localization of the condition difference between expecting an auditory versus a visual target, revealing a significant cluster in early visual areas with stronger effects on the right hemisphere ($p < .01$). C-D, Time-frequency representation of power in the cue-to-target interval (average over electrodes that showed maximal condition difference in A). note: Cluster electrodes are marked in white. Shading represents standard error from the mean; Δ / Σ represents $(a-b)/(a+b)$ normalization.

Replicating our results from study 1, cluster permutation on the Hilbert-transformed alpha activity revealed a significant cluster between -1.47 to -1.18 s ($p = .034$), where alpha change from baseline was higher when expecting an auditory compared to a visual target (see **Fig. 8A**). Applying roughly the cluster time-window (-1.5 and -1 s), we conducted a source localization, contrasting the two conditions with cluster permutation analysis. We found that condition differences in alpha activity originated from early visual areas, albeit with a stronger effect on the right compared to the left hemisphere ($p < .01$; peak spm coordinates: $41 -82 -19$ mm, in the right lingual gyrus see **Fig. 8B**). Additionally, we replicated this effect with a virtual channel analysis in V1 (see **Suppl Fig. 12**).

Cross-modal cues increased the amplitude of the frequency-tagged responses across both modalities

In accordance with study 1, neuronal responses to 40 Hz auditory steady-state evoked field (ASSEF) responses were strongest over temporal areas (**Fig. 9A**) and 36 Hz visual steady-state evoked field (VSSEF) responses were strongest over occipital areas (**Fig. 9C**). As expected, the auditory tagging response originated from the right-hemispheric early auditory cortex (cluster significance: $p < .01$; peak spm coordinates: $69 -22 5$ mm, in the right superior temporal gyrus; see **Fig. 9B**) and the visual tagging response originated from the early visual cortex (cluster significance: $p < .001$; peak spm coordinates: $19 -105 -11$ mm, in the right lingual gyrus; see **Fig. 9D**). Additionally, we observed significantly reduced 40 Hz activity in the left-hemispheric visual-to-central cortex (cluster significance: $p < .01$).

Applying cluster permutation on the Hilbert-transformed frequency-tagging data over the last 500 ms before target onset replicated the results of study 1, where both 36 Hz VSSEF as well as 40 Hz ASSEF signals were significantly enhanced when expecting an auditory compared to a visual target (see **Fig. 9E-G**; auditory target: $p = .043$; visual target: $p = .019$).

Source localization confirmed the condition difference in 36 Hz VSSEF activity to originate from later stages of the processing stream, encompassing a wide range of areas, most strongly the medial occipital cortex and the precuneus (cluster significance: $p = .047$; peak spm coordinates: $3 -66 44$ mm, in the left and right precuneus; see **Fig. 9H**). In source space, the effect was not significant for auditory 40 Hz activity ($p = .11$; peak spm coordinates: $-9 -39 0$ mm, in the left precuneus; see **Fig. 9F**). Furthermore, a virtual channel analysis in V1 and Heschl's gyrus confirmed that there were no condition differences in primary visual and auditory areas (see **Suppl Fig. 12**).

Alpha power was positively correlated with amplitude of frequency tagged responses

The positive correlations between alpha power and frequency-tagging amplitude observed in study 1 could be replicated. Alpha activity 500 ms before target onset correlated both with 36 Hz VSSEF and 40 Hz ASSEF activity when expecting a visual target (see **Fig. 10A-B**, 36 Hz response: cluster significance: $p < .01$; median split: $p < .001$; $t_{(24)} = 4.33$; $BF_{(10)} = 127$; t-test: $p < .001$; $t_{(24)} = 5.33$; $BF_{(10)} = 1272$; 40 Hz response: cluster significance: $p < .001$; median split: $p < .001$; $t_{(24)} = 7.05$; $BF_{(10)} = 59443$; t-test: $p < .001$; $t_{(24)} = 6.75$; $BF_{(10)} = 30515$).

The same correlation was present when expecting an auditory target for both 36 Hz VSSEF activity (see **Fig. 10C-E**; cluster significance: $p < .001$; median split: $p < .001$; $t_{(23)} = 3.62$; $BF_{(10)} = 25.51$; t-test: $p < .001$; $t_{(23)} = 4.60$; $BF_{(10)} = 216$) and 40 Hz ASSEF activity (cluster significance: $p = .005$; median split: $p < .001$; $t_{(25)} = 3.75$; $BF_{(10)} = 36.40$; t-test: $p < .001$; $t_{(25)} = 4.06$; $BF_{(10)} = 73.61$).

In accordance with the results of study 1, we then tested whether alpha activity preceding frequency-tagging activity showed similar correlations. Our data revealed that during the last 1 to 1.5 s before target onset alpha activity correlated with the 36 Hz VSSEF response during the last

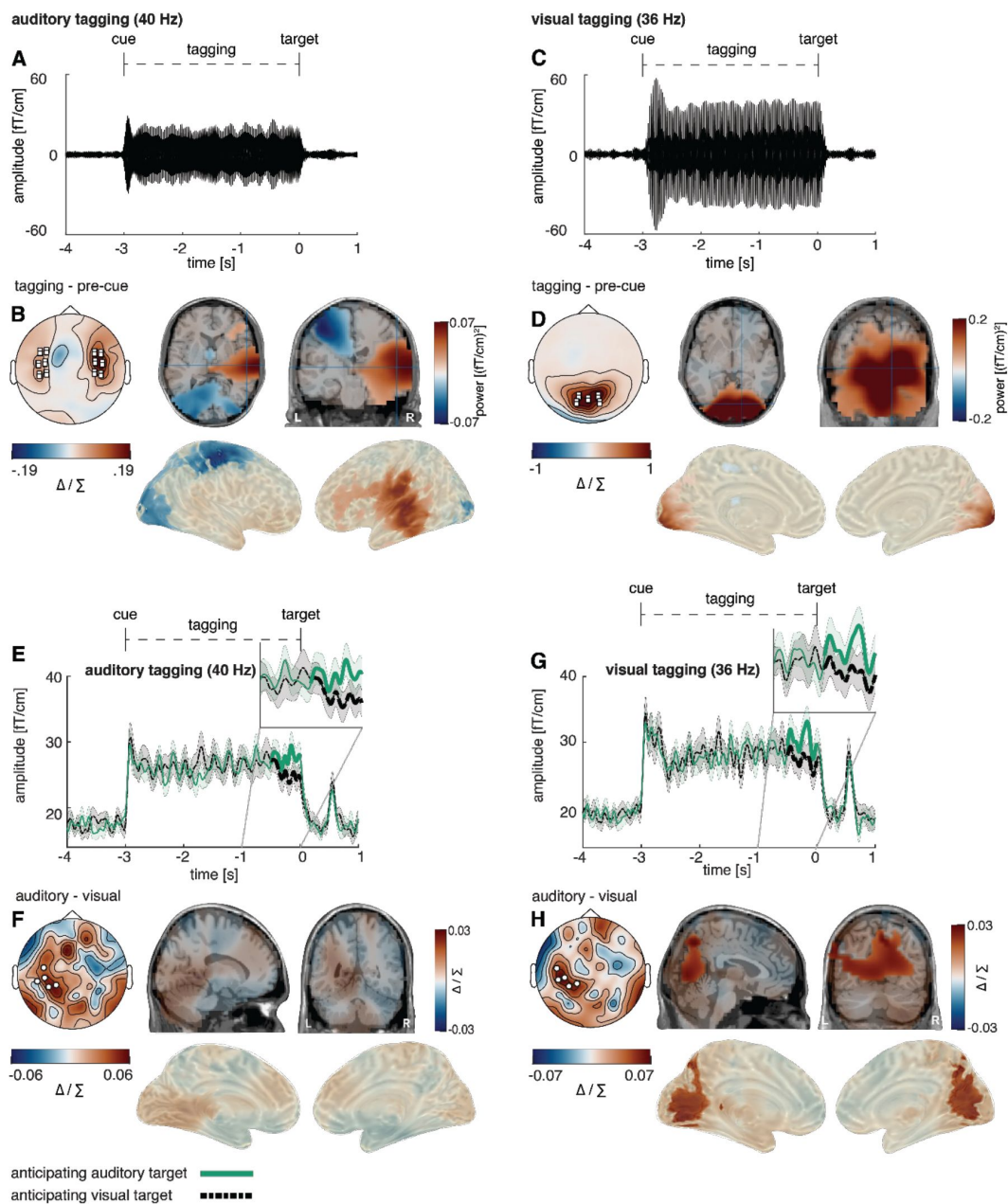


Figure 9.

Post-cue modality specific early visual modulation of alpha power in anticipation of an auditory versus a visual target.

A, auditory steady-state evoked fields (ASSEF) averaged over 24 temporal sensors displaying the highest auditory 40 Hz power (12 right, 12 left; marked in white squares). B, ASSEF source localization revealed a significant positive cluster in the right-hemispheric early auditory cortex ($p < .001$). C, visual steady-state evoked fields (VSSEF) averaged over 10 occipital sensors, displaying the highest visual 36 Hz power (marked in white squares). D, VSSEF source localization revealed a significant positive cluster in the early visual cortex ($p < .001$). E, the Hilbert-envelope of the 40 Hz ASSEF reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p = .043$); F, condition differences in the 40 Hz ASSEF response did not reach significance in sensor space. G, the Hilbert-envelope of the 36 Hz VSSEF likewise reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p = .019$). H, condition differences in the 36 Hz VSSEF response were significant over several areas of the visual stream, including most strongly the medial occipital cortex, the calcarine fissure, and the precuneus ($p = .047$); note: Cluster electrodes are marked in white. Shading represents standard error from the mean. Δ / Σ represents $(a-b)/(a+b)$ normalization.

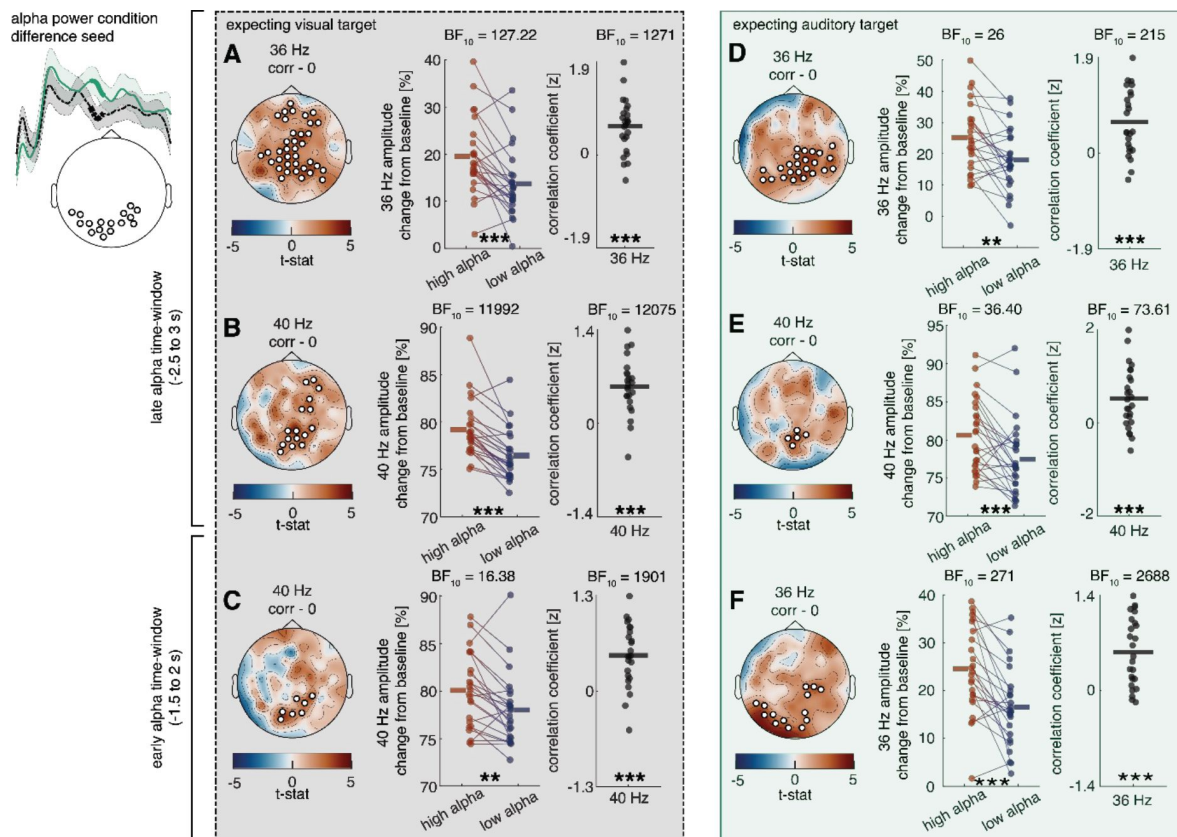


Figure 10.

Relationship between cue induced alpha modulation and amplitude of frequency tagged responses in study 2.

A-C, a positive correlation is visible between alpha activity in the last 500 ms as well as alpha activity in the last 1500ms-1000 ms and steady state potentials shortly before target onset when expecting a visual target (36 Hz late: $p = .013$; 40 Hz late: $p = .009$; 40 Hz early: $p = .002$). D-F, when expecting an auditory target, there is a positive correlation between alpha activity in the last 500 ms as well as alpha activity in the last 1500ms-1000 ms and steady state potentials shortly before target onset (36 Hz late: $p < .001$; 40 Hz late: $p = .005$; 36 Hz early: $p = .011$). $N = 27$; *** sig. $< .001$; ** sig. $< .01$; * sig. $< .05$. + sig. $< .1$.

500 ms prior to an auditory target (see **Fig. 10F**; cluster significance: $p = .01$; median split: $p < .001$; $t_{(24)} = 4.66$; $BF_{(10)} = 271$; t-test: $p < .001$; $t_{(24)} = 5.66$; $BF_{(10)} = 2688$). The same alpha activity correlated with 40 Hz ASSEF activity during the last 500 ms prior to a visual target (see **Fig. 10C**; cluster significance: $p = .002$; median split: $p = .002$; $t_{(23)} = 3.41$; $BF_{(10)} = 16.38$; t-test: $p < .001$; $t_{(23)} = 5.57$; $BF_{(10)} = 1901$).

Lastly, both alpha activity as well as 36 Hz VSSEF responses 500 ms before target onset correlated negatively with reaction time on a trial-by-trial basis, indicating faster reaction times in trials with higher pre-stimulus activity (alpha: $p = .037$; median split: $p = .013$; $t_{(25)} = -2.67$; $BF_{(10)} = 3.78$; t-test: $p < .01$; $t_{(25)} = -3.34$; $BF_{(10)} = 14.84$; 36 Hz: $p = .002$; median split: $p = .004$; $t_{(25)} = -3.20$; $BF_{(10)} = 8.98$; t-test: $p < .01$; $t_{(25)} = -3.46$; $BF_{(10)} = 19.12$. See **suppl Fig. 3-4**).

The same increase in the intermodulatory frequency observed in study 1, could be observed in our second study, during the last 500 ms prior to target onset (see **Fig. 11A**; $p = .006$). In source space, a descriptive condition difference was visible in auditory and visual sensory cortices, however this effect did not reach significance ($p = .49$; peak spm coordinates: 45 -83 -19 mm, in the right lingual gyrus; see **Fig. 11B**).

Discussion

The neuropsychological account of attention defines it as the selective facilitation (i.e., prioritization) of relevant sensory input and suppression of irrelevant sensory input. Oscillatory activity in the alpha range (~10 Hz) has been suggested to play a mechanistic role in attention through inhibition of irrelevant cortices, commonly referred to as the ‘alpha inhibition hypothesis’ (Foxe et al., 1998; Jensen & Mazaheri, 2010; Klimesch et al., 2007). In the current cross-modal attention study we directly tested this hypothesis by using frequency-tagging to specifically examine how cues signalling the modality of an upcoming target (either the auditory or visual modality) affected the responsiveness of the relevant and irrelevant sensory cortices prior to target onset. In-line with previous work, we observed a post-cue increase in posterior alpha power in anticipation of processing auditory targets. However, contrary to prevalent theories proposing visual gain suppression when focusing on the auditory modality, we observed that the amplitude of visual frequency-tagging responses increased just prior to the onset of the auditory target. This suggests that responsiveness of the visual stream was not inhibited when attention was directed to auditory processing and was not inhibited by occipital alpha activity, which directly contradicts the alpha inhibition hypothesis. Our results reconcile previously paradoxical results on audio-visual attention and support the view that alpha activity gates downstream communication pathways.

Frequency-tagging

In the current experiment, we specifically chose to analyse the cue-to-target interval, where both visual and auditory SSEPs/SSEFs present preparatory states for the upcoming task, independent of task-related processing or performance. The magnitude of auditory SSEPs/SSEFs was increased shortly before target onset when expecting an auditory target compared to a visual target, very much in line with previous reports (e.g., Saupe et al., 2009).

In contrast to the results reported in Saupe et al., (2009), where visual SSEPs decreased when attending the auditory modality, visual SSEPs/SSEFs increased shortly before target onset when expecting an auditory target in our data. This is especially surprising as auditory targets were frequently, or in case of our second study, always accompanied by visual distractors, rendering it optimal for task success to completely ignore any visual input. As auditory targets were significantly more difficult than visual targets in our first study and of comparable difficulty in our second study, these results strongly speak to a vigilance increase of sensory processing

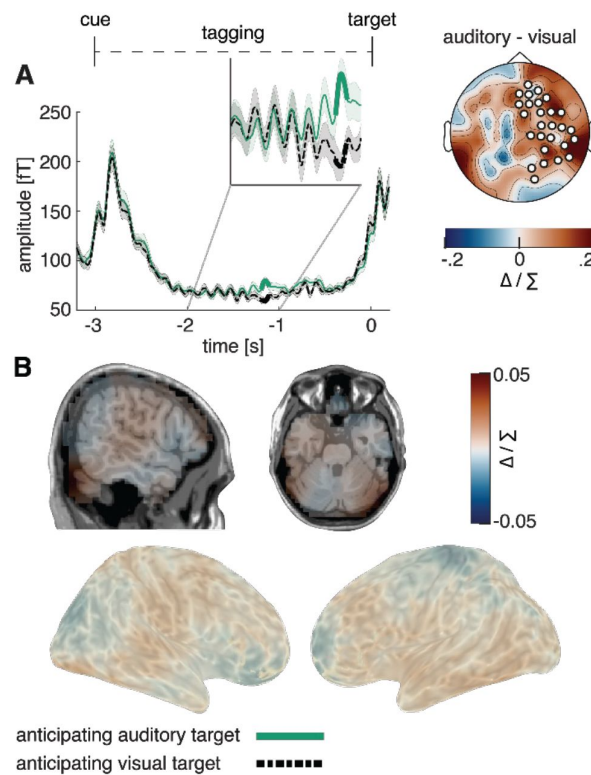


Figure 11.

Steady-state response in the intermodulation frequency.

A, similar to study 1, 4 Hz frequency-tagging activity was increased shortly before target onset when expecting an auditory compared to a visual target ($p = .006$). D, source localization showed activity over auditory sensory areas, but did not reach significance. $N = 27$.

independent of modality and an inability to selectively disengage one sensory modality in anticipation of a demanding task. This view is consistent with previous work in which visual SSEPs elicited by irrelevant background stimulation increased with task load in an auditory discrimination task (Jacoby et al., 2012). Furthermore, our results indicate that task demand is a strong candidate to reconcile previously seemingly paradox results, as splitting attention between the auditory and visual system seemed possible in simpler tasks (Driver & Spence, 1998; Saupe et al., 2009) and impossible under high demand (de Jong et al., 2010; Driver, 1996; Driver & Spence, 1998; Spence & Driver, 1996). An alternative account for our findings stems from the evidence, that participants are more likely to only perceive and react to the visual modality, when confronted with audio-visual stimuli (Colavita, 1974; Spence, 2009). However, this effect was mostly limited to speeded modality discrimination/target detection tasks (Sinnott et al., 2008; Spence, 2009). Furthermore, the increased difficulty of the here-used auditory stimuli was confirmed in a previous block-design study (van Diepen & Mazaheri, 2017) and in our second study, performances over the visual and auditory tasks were comparable. Nevertheless, visual dominance could play a role for auditory target difficulty as well as predictions over the reciprocity of the audio-visual relationship.

A revision of the alpha inhibition hypothesis

Top-down cued changes in alpha power have now been widely viewed to play a functional role in directing attention: the processing of irrelevant information is attenuated by increasing alpha power in cortices involved with processing this information (Foxe, Simpson, & Ahlfors, 1998; Hanslmayr et al., 2007; Jensen & Mazaheri, 2010). However, recent evidence suggests that posterior alpha activity does not inhibit gain in early sensory processing stages (Antonov et al., 2020; Gundlach et al., 2020; Gutteling et al., 2022; Zhigalov & Jensen, 2020). To date there has been no direct investigation into the effect of alpha increases on later stages in the processing stream. In the current study, as expected, we observed a post-cue increase in occipital alpha activity in anticipation of an auditory target. However, we also observed an increase in the amplitude of visual SSEPs during the cue-target interval, which directly contradicts the widespread view of alpha activity exerting ‘gain control’ in early sensory areas by regulating excitability (Foxe & Snyder, 2011; Jensen & Mazaheri, 2010; Van Diepen et al., 2019). Here we propose that alpha activity, rather than modulating early primary sensory processing, exhibits its inhibitory effects at later stages of the processing stream (Antonov et al., 2020; Gundlach et al., 2020; Zhigalov & Jensen, 2020; Zumer et al., 2014), gating feedforward or feedback communication between sensory areas (Bauer et al., 2020; Haegens et al., 2015; Uemura et al., 2021). Our data provides evidence in favour of this view, as we can show that early sensory alpha activity covaries over trials with SSEP magnitude in higher order sensory areas. If alpha activity exerted gain control in early visual regions, increased alpha activity would have to lead to a decrease in SSEP responses. In contrast, we observe that increased alpha activity originating from early visual cortex is related to *enhanced* visual processing. Source localization confirmed that this enhancement was not originating from early visual areas, but from areas associated with later stages of the processing stream such as the precuneus, which has been connected to sensory integration (Al-Ramadhani et al., 2021; Xie et al., 2019). While we cannot completely rule out alternative explanations, it seems plausible to assume that inhibition of other task-irrelevant communication pathways leads to prioritised and thereby enhanced processing over relevant pathways. In line with previous literature (Morrow et al., 2023; Peylo et al., 2021; Zhigalov & Jensen, 2020), we therefore suggest that alpha activity limits task-irrelevant feedforward communication, thereby enhancing processing capabilities in relevant downstream areas (see Fig. 1A). The benefit of alpha-modulation and its effect on visual information processing is underlined by reaction times, which were faster both for trials with high pre-target alpha activity and high pre-target visual SSVEF activity.

It should be noted, the comparison between modulation in alpha activity and in SSEP/SSEFs is difficult, especially concerning timing. This is largely owed to differences in signal-to-noise due to trial averaging in the frequency versus the time domain and temporal and frequency lag in the

estimation of alpha activity (Peylo et al., 2021 [↗](#)). It is further noteworthy, that the majority of evidence for the alpha inhibition hypothesis focused on the effect of pre-target alpha modulation on behaviour and target-related potentials (Morrow et al., 2023 [↗](#)). However, in our data alpha modulation occurs clearly ahead of SSVEP/SSVEF modulation on a scale that could not be simply explained by temporal or frequency smearing. Additionally, significant trial-by-trial correlations, which occur in the frequency domain for both signal types, underline the strong relationship between both measurements.

Interestingly, we could show that the magnitude of the correlation between alpha power and visual information processing varied between conditions, suggesting a dynamic and adaptive regime. This notion supports the view that alpha oscillations represent a mechanism rather than a specific function, which can fulfil different roles depending on task demand and network location, which has been confirmed in a recent study revealing functionally distinct alpha networks (Clausner et al., 2024 [↗](#)). As such, it is conceivable that alpha oscillations can in some cases inhibit local processing, while in other cases, depending on network location, connectivity and demand, alpha oscillation can facilitate signal transmission. In different contexts, utilizing unimodal targets and distractors, spatial cueing, or covert attention, different functional processes could be involved (Morrow et al., 2023 [↗](#)). Future research should intensify efforts to disentangle these effects, investigating localized alpha networks intracranially or through combinations of fMRI, EEG and MEG, to clearly measure their effects on sensory processing and behaviour.

It is known that the localisation of alpha activity reflects the retinotopic organisation of visual spatial attention topographically over the parietooccipital cortex (Kelly et al., 2006 [↗](#); Popov et al., 2019 [↗](#)). Notably, recent studies provided evidence, that the same organisation can be observed for auditory attention. Specifically, the localisation of visual alpha activity in the parietooccipital cortex reflects the spatial direction of auditory attention (Klatt et al., 2021 [↗](#); Popov et al., 2021 [↗](#)). This observation can be explained through micro-saccades towards the spatial location of sounds, which are irrevocably connected to alpha oscillations (Popov et al., 2021 [↗](#)). While we did not manipulate spatial attention, our results fit well to the notion of visual alpha activity serving as a sensory orientation system, relaying visual information to task-relevant downstream processing areas, and blocking communication to irrelevant pathways.

The intermodulation frequency

Previous research showed that simultaneous frequency-tagging in multiple frequencies evokes a response in the intermodulation frequency ($f_1 - f_2$). In multimodal settings, this frequency is thought to reflect cross-modal integration (Drijvers et al., 2021 [↗](#)). This is very well in line with our findings, where increased vigilance of the sensory system arising from anticipation of a difficult auditory target resulted in an increase in the intermodulation frequency. Likewise, our data shows that visual signal enhancement was localized in precuneus, which has been connected to sensory integration (Al-Ramadhani et al., 2021 [↗](#); Xie et al., 2019 [↗](#)). Furthermore, we could show that the intermodulation frequency covaries over trials with reaction time in the easiest condition, where visual targets were presented without any distractors. A lack of this connection in other conditions might reflect increasing interferences from higher task difficulty, rather than a lack of the effect itself, but this remains to be tested. We cannot exclude an alternative explanation, as theta oscillations are known to be involved in movement preparation, it is possible that phase-resets could lead to time-locked appearance of these oscillations (Lakatos et al., 2008 [↗](#); Tomassini et al., 2017 [↗](#)).

Conclusion

Our results taken together suggest that under high attention, audio-visual excitability is enhanced, reflecting an increase in vigilance for both visual as well as auditory information, even if this increases processing of distracting information. We showed that this vigilance shift, as reflected by

SSEP/SSEF responses, is regulated by early visual alpha activity, presumably through relaying of visual information over communication pathways, thereby controlling the downstream flow of visual information.

Materials and methods

Participants EEG-Study

In total, 24 healthy volunteers participated in this study (mean age: 19.1 ± 1.8 SD; 17 women). Due to technical difficulties, one participant could not finish the experiment and one participant did not exceed chance level in the behavioural task ($\sim 33\%$). Both were therefore removed from any further analysis. All remaining participants reported normal or corrected-to-normal vision, no history of psychiatric or neurological illness and provided written informed consent. After completion of the experiment, participants received either monetary compensation or certification of their participation for their university course program. The study protocol was approved by the Ethics Committee of the School of Psychology at the University of Birmingham and is in accordance with the Declaration of Helsinki.

Participants MEG-Study

In total, 28 healthy volunteers participated in this study (mean age: 23.4 ± 3.6 SD; 20 women). One participant was removed from further analysis, as they only responded to $\sim 42\%$ of trials correctly in the second block, which related to 27/19 trials per condition respectively. All remaining participants reported normal or corrected-to-normal vision, no history of psychiatric or neurological illness and provided written informed consent. After completion of the experiment, participants received either monetary compensation or certification of their participation for their university course program. The study protocol was approved by the Ethics Committee of the School of Psychology at the University of Birmingham and is in accordance with the Declaration of Helsinki.

Cross-modal attention paradigm EEG-Study

Each trial was initiated by a brief presentation of a cue (100 ms) signalling the modality of the upcoming discrimination task (v-shape: visual modality; inversed v-shape: auditory modality; diamond-shape: non-specific). During the following cue-to-target interval, the fixation cross was frequency-tagged at 36 Hz. At the same time, a 40 Hz frequency-tagged sound (amplitude modulated white noise) was played over headphones (see [Fig. 1](#)). The volume of tones was initially adjusted to a level that was clearly perceivable but not uncomfortable and remained stable over participants. No task was connected to this interval and participants did not need to pay attention to either the sound or the fixation cross. After three seconds, the frequency-tagging stopped, and right after the cessation of stimuli, the target was presented for a very brief moment (25 ms). It consisted either of a Gabor patch (visual modality) or a sound (auditory modality). If the target modality was visual, participants had to use the three arrow buttons on the keyboard to indicate whether the Gabor patch was tilted to the left (-10° ; left arrow button), vertical (0° ; down arrow button), or tilted to the right (10° ; right arrow button). Additionally, in 50% of the trials, a random distractor from the pool of auditory targets was presented simultaneously to the visual target over headphones. Similarly, if the target modality was auditory, participants used the same buttons to indicate whether the pitch of the tone was low (500 Hz), medium (1000 Hz) or high (2000 Hz). Again, in 50% of the trials, a random distractor from the pool of visual targets was also presented. If the cue was non-specific (diamond shape), either a visual (50% of non-specific trials) or an auditory target was presented, never accompanied by any distractors. Experimental trials were separated by an inter-trial interval of 4 seconds, to avoid carry-over effects from previous trials. The resulting 6 conditions were randomly ordered and balanced out over the experiment.

During the experiment, participants were instructed to keep their gaze locked to a fixation cross presented at the centre of the screen. Preceding data collection, participants performed 36 practice trials to get accustomed to the task and the target stimuli. The ensuing experiment was split into 26 trial sequences, separated by self-chosen breaks, which together resulted in 468 trials and lasted between 80 and 90 minutes. The discrimination task was programmed and presented with MATLAB® R2020b and Psychtoolbox-3 on an LCD-monitor featuring a 140 Hz refresh rate. The onset of the visual and auditory tagging frequencies (i.e. steady state stimuli) were tracked using the Cedrus Stimtracker (<https://cedrus.com/stimtracker/index.htm> .

Adjustments to the attention paradigm in the MEG-Study

In our second study, we removed all ambiguity concerning targets and distractors and therefore developed a blocked design, incorporating two blocks. The first block did not display distractors and only cues which correctly predicted the target were presented. Cues in the second block were likewise always correctly indicating the target modality, but this time, each target was accompanied by a random distractor from the non-target modality.

Furthermore, the visual task was adjusted to be more in line with the auditory task. As such, the Gabor patches now featured stripes in different frequencies (e.g. a low number of stripes, a medium number of stripes and a high number of stripes). The participant's task was to discriminate between these three Gabor patches. As auditory targets had been markedly more difficult in our first study, we now included a brief difficulty calibration prior to the experiment. First, we presented 21 Gabor patches with 3 different amounts of stripes following a standard difficulty. If participants could discriminate them correctly 55 – 75% of the time, this difficulty setting was chosen. Otherwise, depending on the performance, the stripe-frequency of the Gabor patches was adjusted. There were maximally 3 sessions of 21 Gabor patches, after which we had enough data to calibrate the individual difficulty setting.

The same procedure was then performed with the difficulty of the tones, calibrating the pitch frequency for each individual participant.

Lastly, visual frequency-tagging stimulation now followed a sinusoidal contrast-change rather than an on-off stimulation, which was possible due to a high-resolution projector featuring a refresh rate of 1440 Hz (PROPixx DLP LED projector; VPixx Technologies Inc., Canada).

Eye-tracking

To confirm that participants had focused on the fixation cross during the cue-to-target interval, we incorporated eye-tracking into our MEG-experiment (EyeLink 1000 Plus). Correct trials of the second block were analysed for vertical and horizontal eye-movements. To exclude blinks from this analysis, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the eye-tracking data (See **suppl Fig. 5** .

Behavioral analysis

We were interested in accuracy in the discrimination of visual targets and auditory targets, as well as reaction times. Furthermore, we examined the distraction cost of having a target presented with a distractor of a different modality as well as the reaction time to make the target discrimination. The distraction cost was calculated as the reaction time difference between cued targets with distractors (i.e. visual and auditory stimuli presented together) and cued targets without distractors (either a visual or auditory stimulus presented alone). All incorrect trials as well as trials with reaction times faster than 100 ms or exceeding 1500 ms were removed from analysis (0.5 % too fast, 9.2% too slow).

EEG data acquisition

All EEG recordings were conducted using a WaveGuard Cap (ANTneuro), featuring 64 Ag/AgCL electrodes (10-10 system; ground: Fz; reference: Cpz; EOG: left canthus). Electrodes positions were prepared with OneStep cleargel conductive paste and impedances were kept below 100 k Ω . The measured signal was transmitted using an ANTneuro EEGosports amplifier (low-pass filter: 150 Hz; high-pass filter: 0.5 Hz; sampling rate: 500 Hz).

MEG data acquisition

Prior to the experiment, fiducial positions and head-shape were recorded using a FASTRAK system (Polhemus, USA). The experiment took place in a dimly lit room, where participants were seated in a comfortable chair in the gantry of a 306-sensor TRIUX Elekta system with 204 orthogonal planar gradiometers and 102 magnetometers (Elekta, Finland). The 71*40 cm screen was positioned at ~1.40 m distance from the participant.

EEG Preprocessing

Offline analyses were performed in MATLAB ® R2020b. The data was pruned from artifacts by visual inspection using the EEGLAB toolbox (Delorme & Makeig, 2004). Additionally, blinks and ocular artefacts were removed from the data using independent component analysis (ICA). EEG channels were re-referenced to an average of all channels (excluding EOG).

MEG Preprocessing

Offline analyses were performed in MATLAB ® R2020b and Python. Spatiotemporal Signal-Source-Separation (SSS) was applied to the raw data via MNE's inbuilt maxfilter function with a duration window of 10 s and a correlation value of .9. The data was pruned from artifacts by visual inspection using the Fieldtrip toolbox (Delorme & Makeig, 2004). Additionally, blinks and ocular artefacts were removed from the data using independent component analysis (ICA). In sensor space, planar gradiometers were combined for further analyses. In source space, all individual planar gradiometers were analysed.

Amplitude of the evoked frequency-tagging response

To investigate the temporal dynamics of amplitude of the frequency tagged responses after the onset of the attentional cues (also the precise onset of the frequency tagged stimuli) the data was epoched into 6-second segments starting 1.5 seconds prior to cue onset. Next the data were narrow-band filtered around the 36 Hz activity to capture the visual frequency-tagging, 40 Hz activity in to capture the auditory frequency-tagging, and the intermodulation frequency at 4 Hz, which can be derived by subtracting both frequency-tagging responses ($f_i = f_{\text{auditory}} - f_{\text{visual}}$; see Drijvers, Spaak & Jensen, 2020). Here we used a Blackmann-windowed sinc filters adapted to a suitable ratio of temporal and frequency resolution for the specific frequency of each of the tagged signals: filter order 116 for 35.5 to 36.5 Hz and 39.5 to 40.5, filter order 344 for 3.5 to 4.5 Hz. The filtered data at each of the tagged frequencies as well the intermodulated frequency were baseline corrected (interval between 700 and 200 ms preceding cue onset) before calculating the average over trials to obtain steady-state evoked potentials). The power envelope of the SSEPs of tagged frequencies was estimated using Hilbert transformation. To confirm that 4 Hz is a sufficient distance between tagging frequencies, we repeated to analysis for 43.5 to 44.5. We found no indication of frequency-bleeding over, as the effects observed at 40 Hz, were not present at 44 Hz (see **Suppl Fig. 11** [↗](#)).

Temporal dynamics of the induced EEG changes

In addition to looking at the cue evoked changes in the amplitude of the frequency tagged signals, we investigated the induced changes in the EEG signal at the frequencies of the tagged auditory and visual stimuli, alpha activity (9-11 Hz, filter order 276 for 7.5 to 12.5 Hz), as well as the intermodulation frequency (4 Hz; filter order 344 for 3.5 to 4.5 Hz). Here rather than averaging the epoched data filtered at the specific frequency ranges, we performed the Hilbert transform, and averaged the power-envelope of the specific frequencies across trials. This approach is very much analogous to the standard time-frequency analysis using convolutions ([van Diepen & Mazaheri, 2017](#); [Zhigalov et al., 2019](#)), but affords more control concerning temporal versus frequency resolution to examine the temporal dynamics of the specific frequencies of interest. In our second study, the individual peak alpha frequency was used for bandpass-filtering in contrast to a standardised band applied in the first study.

Time-frequency representations of power

In addition to the estimating frequency power envelopes, Time-frequency representations (TFRs) of power of the EEG signal were estimated using the Fieldtrip toolbox (Oostenveld et al., 2011). The power or frequencies between 5 and 20 Hz were calculated for each trial, using a sliding time window (frequency steps: 0.5; time steps: 10 ms). The length of the window was adjusted to a length of 3 cycles per frequency and tapered with a Hanning window. For each trial, both datasets were normalised to display relative percent change from baseline using the following formula: $[(\text{activity} - \text{baseline}) / \text{baseline}]$, where baseline refers to the interval between 700 and 200 ms before cue onset. To estimate the topographical distribution of voltage differences between conditions, uncorrected power values were normalised applying the following formula: $[\Delta / \Sigma = (a - b) / (a + b)]$, where a and b reflect the different conditions.

Source localization Source localization was performed with a beamformer approach using the Fieldtrip toolbox. Headmodels were created based on individual T1-scans fitted to fiducial points and head shapes. These data were fit to a 5mm 3d sourcemodel and warped into MNI-space. Two participants were missing individual T1-scans. In these cases, we applied a standardized T1-scan using the Colin 27 Average Brain Model ([Holmes et al., 1998](#)). Frequency-domain data was localized using the Dynamic Imaging of Coherent Sources (DICS beamformer) method with a dpss taper of 2 Hz assuming fixed orientation. As condition differences between frequency-tagging responses were better estimated in the time-domain, we assessed them applying the Synthetic Aperture Magnetometry (SAM-beamformer) method with optimal fixed rotation ([Sekihara et al., 2004](#)). Significant brain areas and peak coordinates were related to brain areas using the Anatomical Automatic Labeling (AAL) atlas for SPM8 ([Tzourio-Mazoyer et al., 2002](#)). After statistical analysis, source localized data was interpolated onto the Colin 27 Average Brain Model MRI. Cerebellar and brainstem interpolations were excluded from the coordinate system.

Statistical analysis

Condition differences in the behavioural task were estimated with paired t-tests and repeated measures ANOVAS, utilizing Tukey-Kramer Post-Hoc test. Power differences between conditions, as well as source-space contrasts were analysed using cluster permutation analysis ([Maris & Oostenveld, 2007](#)). In this procedure, condition labels were randomly shuffled 1000 times, creating pairs of surrogate conditions. To test for significance, paired t-tests were conducted for each data point and each channel, resulting in one t-matrix for real conditions and 1000 t-matrices for surrogate conditions. Significant t-values ($p < .05$) were defined as clusters if there was at least one significant data point present at the same time and frequency in at least two neighbouring channels. To correct for multiple comparisons, a condition difference was only assumed, if the maximum sum of t-values in a real cluster exceeded the same sum of 95% of the clusters found in the surrogate data. To replicate our results in the second study, we applied the same statistics and

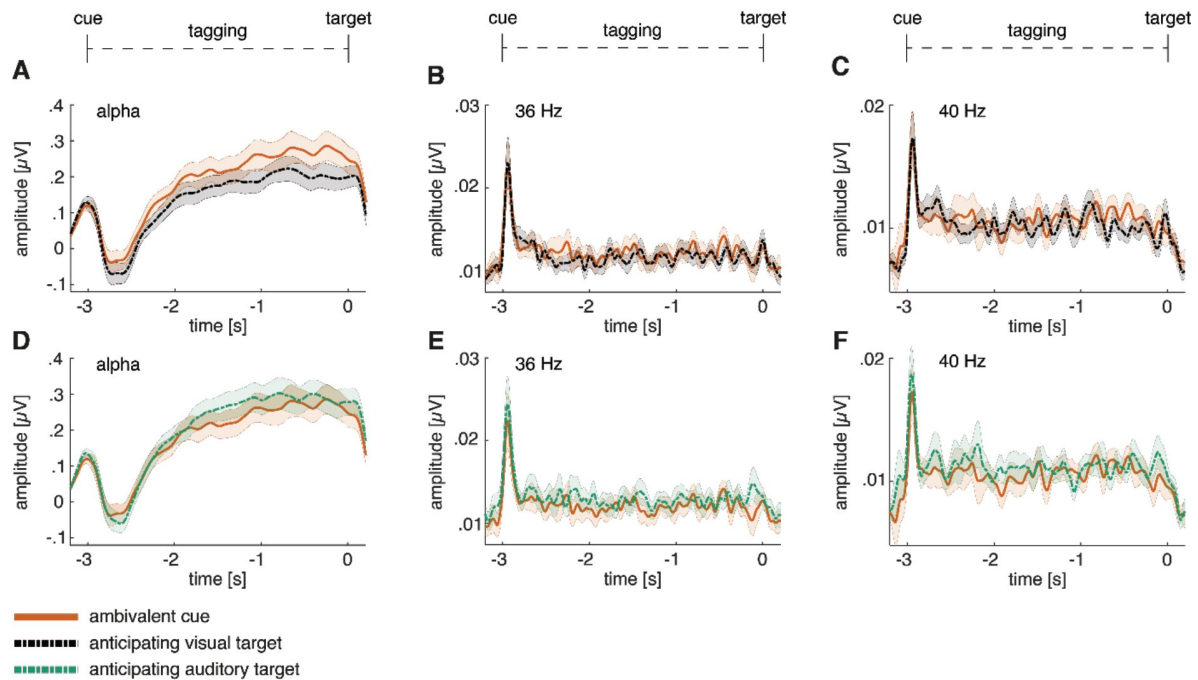
averaged the previously found time-intervals into windows of 500ms (e.g.: if we found an effect -0.51 to -0.0620 s prior to target onset we tested this effect for a time window of -0.5 to 0 s prior to target onset).

The relationship between induced changes in alpha activity and frequency tagged responses was assessed using trial by trial Spearman correlations. For each participant and each electrode, a correlation coefficient was calculated between the average activity in a previously identified cluster, which was used as seed (e.g., condition differences in alpha activity), and the average activity of the electrophysiological correlate of interest (e.g., 36 Hz activity over the previously identified time window). The correlation coefficients were z-transformed, and the resulting channel by participant matrix was tested against null-correlation model using the cluster permutation approach described above. Derived clusters were additionally tested and visualised by comparing median split trials of high vs low activity. For this analysis, outliers (values deviating more than 2 standard deviations from the mean) were excluded. Furthermore, the average correlation coefficient of the cluster was tested against a 0-correlation model for each participant using t-statistics. Lastly, an interaction between electrophysiological correlations and conditions was performed using correlation coefficients for each participant and electrode, testing them between conditions using the cluster permutation approach. Perceptually uniform and universally readable colormaps were applied to all visualisations (Cramer et al., 2020 [DOI](#)). All data are presented as mean $\pm$ standard error of the mean (SEM).

Data availability

Codes for analyses and figures will be made available on github and the data will be uploaded to Dryad or OSF

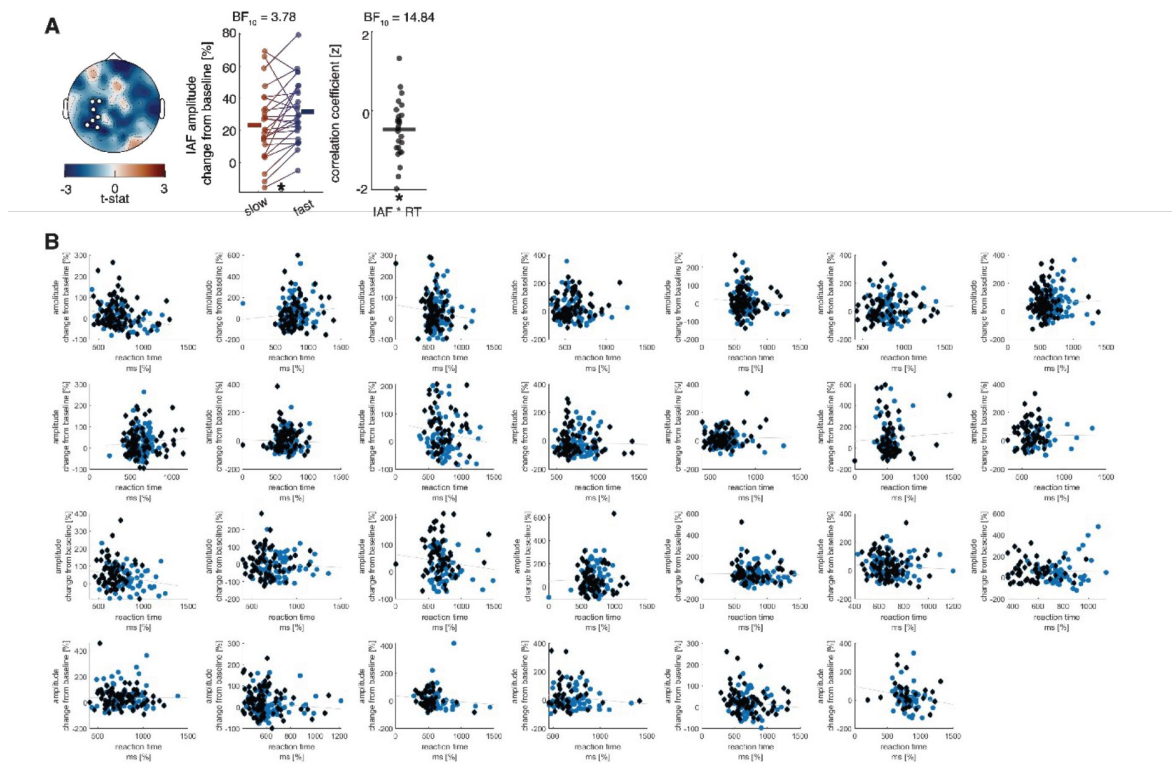
Supplementary figures



Suppl Fig. 2.

Time course of alpha activity and frequency-tagging responses for the non-specific compared to the visually-cued condition.

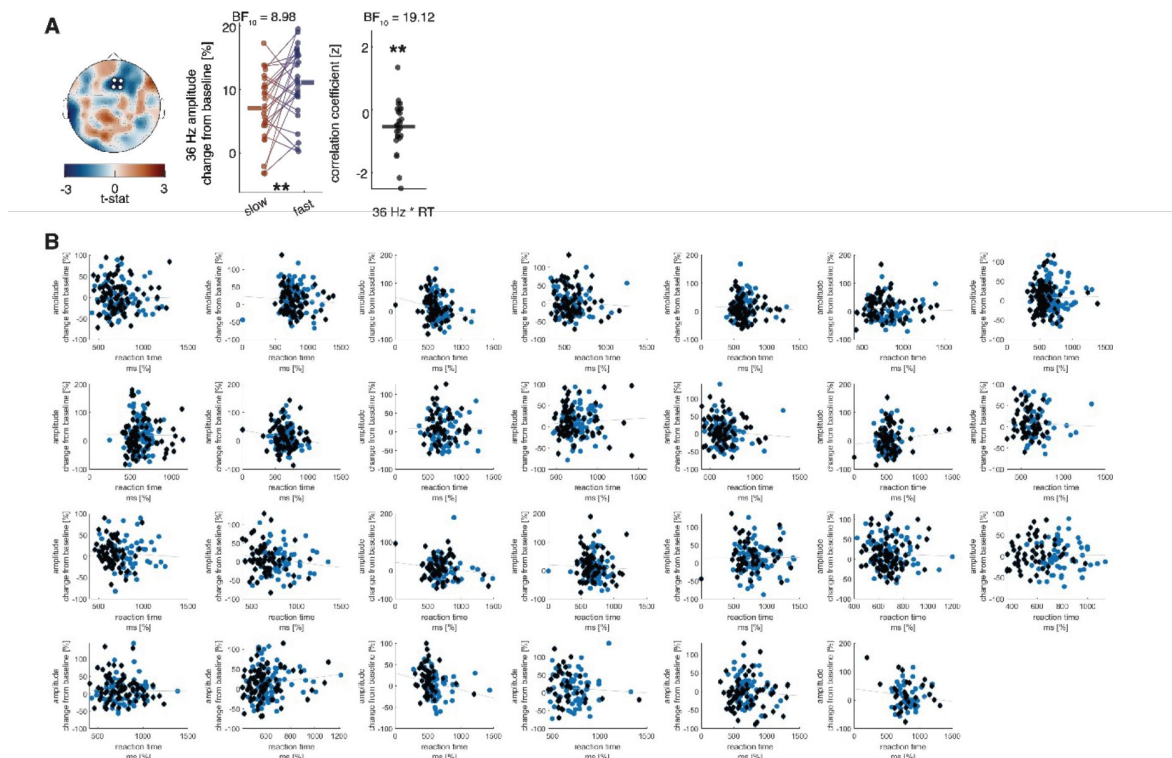
A, alpha activity compared between expecting a visual target and having received a non-specific cue. B, 36 Hz frequency-tagging response between expecting a visual target and having received a non-specific cue. C, 40 Hz frequency-tagging response between expecting a visual target and having received a non-specific cue. D, alpha activity compared between expecting an auditory target and having received a non-specific cue. E, 36 Hz frequency-tagging response between expecting an auditory target and having received a non-specific cue. F, 40 Hz frequency-tagging response between expecting an auditory target and having received a non-specific cue.



Suppl Fig. 3.

Correlation of prestimulus alpha change from baseline with reaction time in the MEG study.

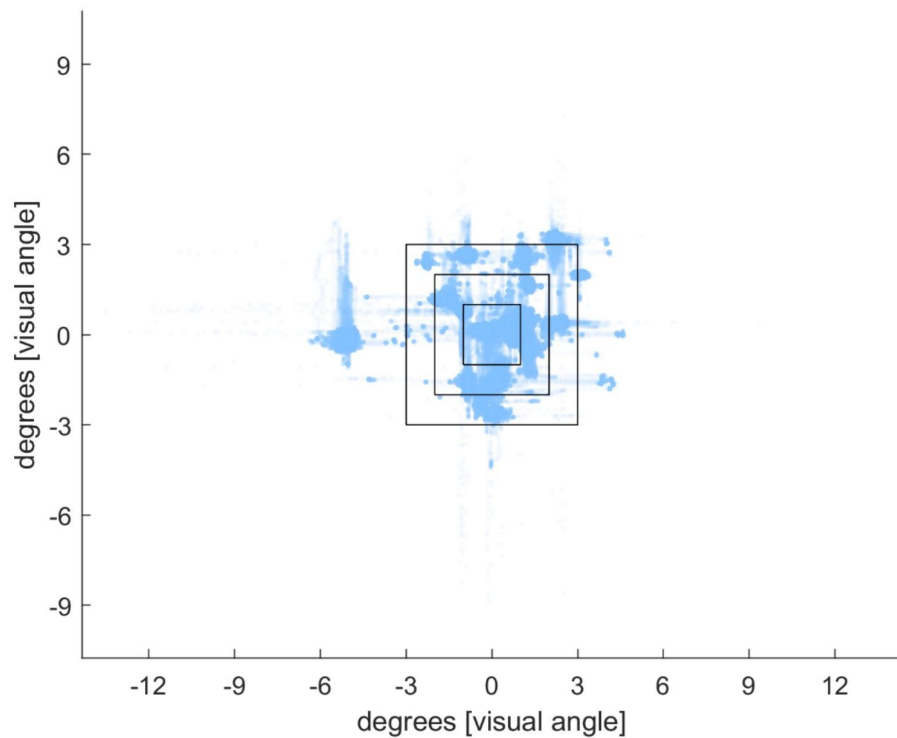
A, the analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically ($p = .037$). Additionally, median splits between fast and slow reaction time trials ($p = .013$; $t_{(25)} = -2.67$) as well as correlation coefficients ($p = .003$; $t_{(25)} = -3.34$) of these clusters are displayed for all participants. A negative correlation is visible between alpha modulation and reaction times in the last 500 ms before target onset when expecting a visual target. B, Correlation between alpha modulation and reaction time for each participant. Black diamonds represent trials from the first block (without distractor) and blue dots represent trials from the second block (with auditory distractor).



Suppl Fig. 4.

Correlation of 36 Hz change from baseline with reaction time in the MEG study.

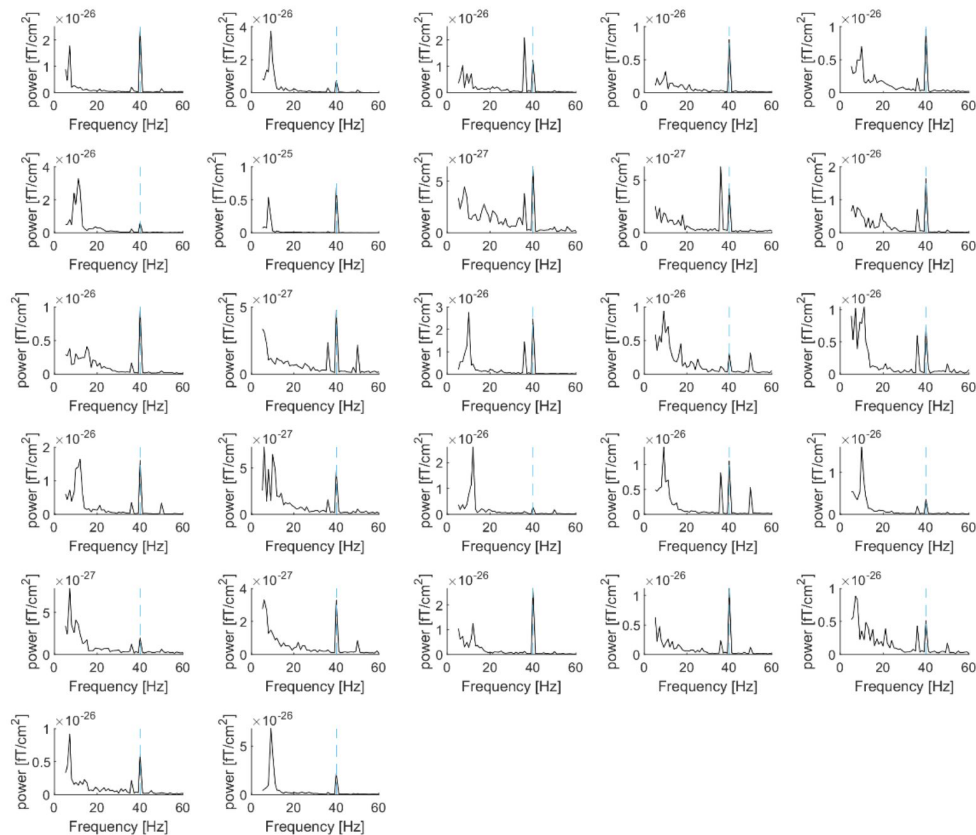
A, the analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically ($p = .040$). Additionally, median splits between fast and slow reaction time trials ($p = .005$; $t_{(25)} = -3.10$) as well as correlation coefficients ($p = .002$; $t_{(25)} = -3.46$) of these clusters are displayed for all participants. A negative correlation is visible between 36 Hz modulation and reaction times in the last 500 ms before target onset when expecting a visual target. B, Correlation between 36 Hz modulation and reaction time for each participant. Black diamonds represent trials from the first block (without distractor) and blue dots represent trials from the second block (with auditory distractor).



Suppl Fig. 5.

Illustration of eye-tracking during the cue-to-target interval (2.5 – 0 s before target onset).

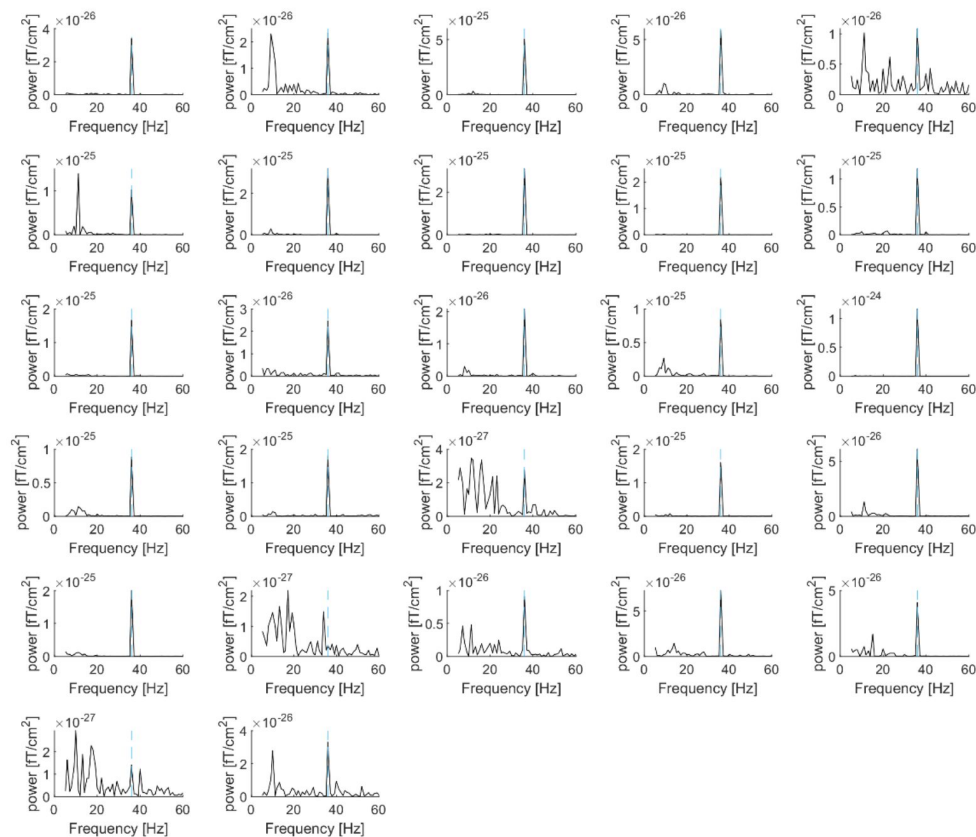
All datapoints of eye-positions during the cue-to-trial interval for all trials and all participants were plotted with 5% visibility on top of each other. Only 3% of datapoints showed eye-movement larger than 3 degrees of visual angle away from the fixation cross.



Suppl Fig. 6.

Individual ERP power spectra of the cue-to-target interval when anticipating an auditory target in the MEG-study.

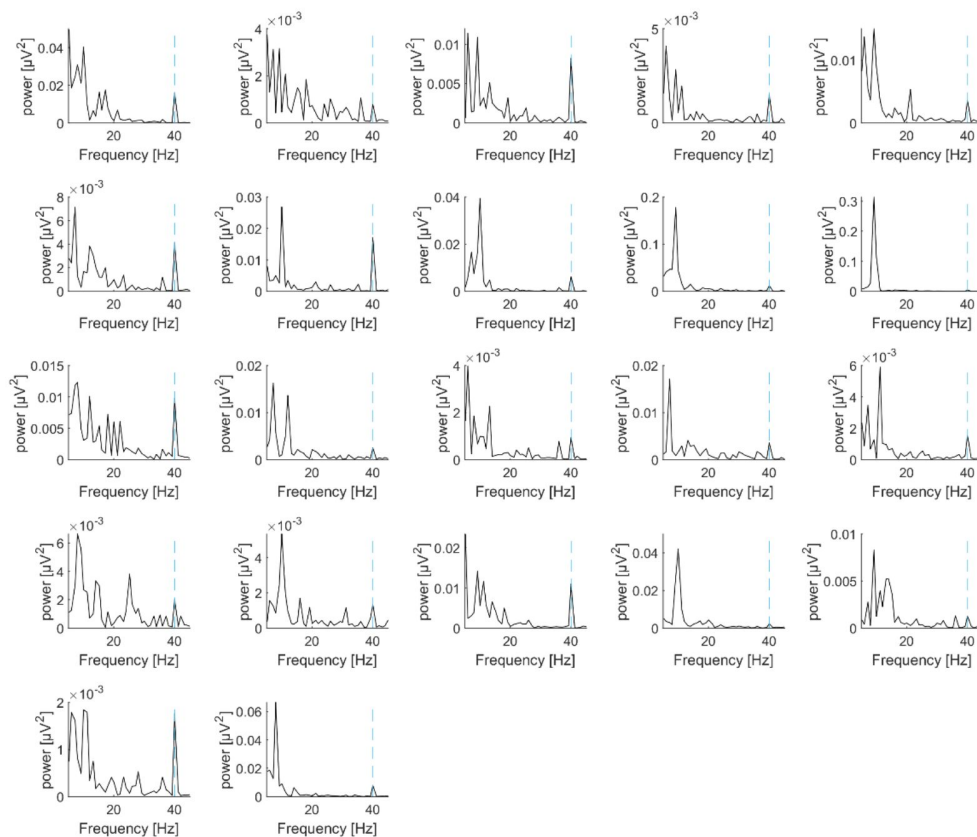
Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 40 Hz (auditory frequency-tagging).



Suppl Fig. 7.

Individual ERP power spectra of the cue-to-target interval when anticipating a visual target in the MEG-study.

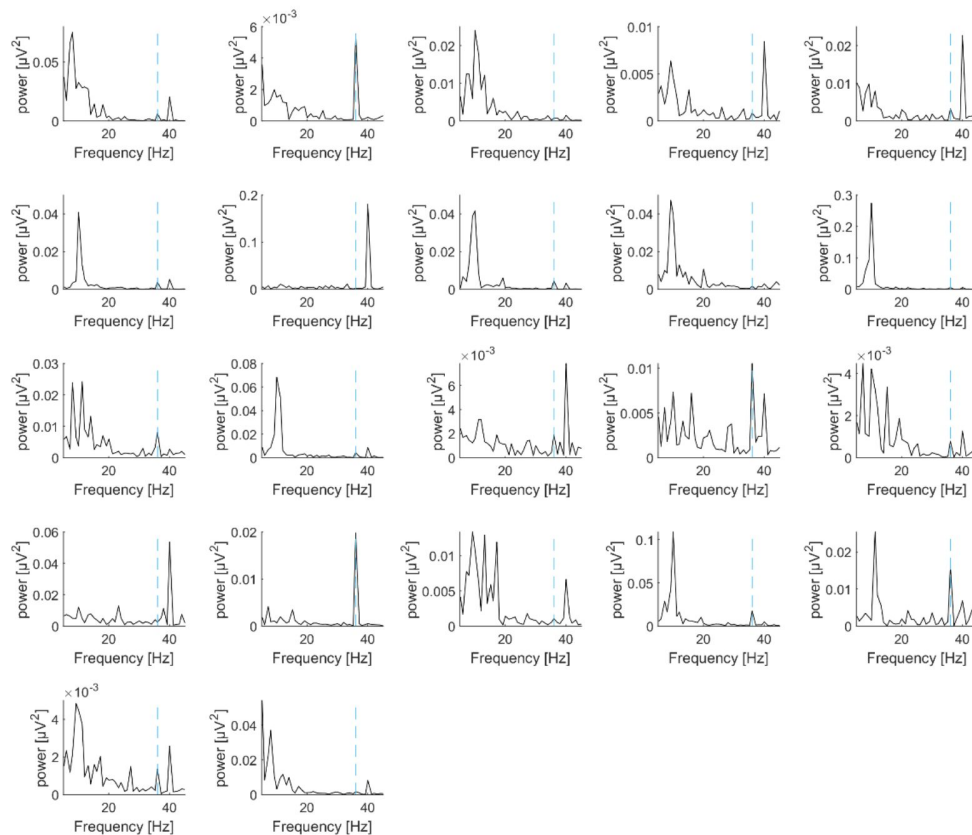
Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 36 Hz (auditory frequency-tagging).



Suppl Fig. 8.

Individual ERP power spectra of the cue-to-target interval when anticipating an auditory target in the EEG-study.

Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 40 Hz (auditory frequency-tagging).



Suppl Fig. 9.

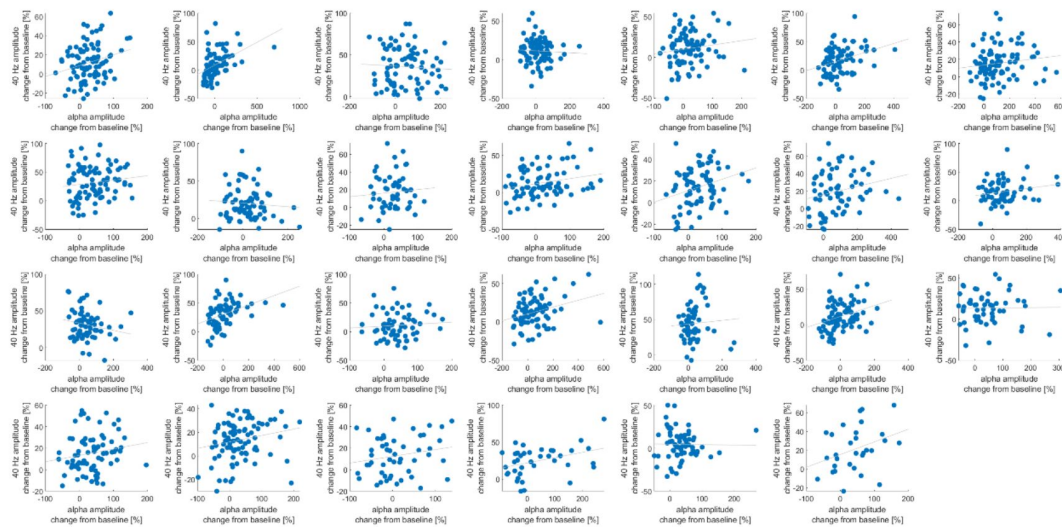
Individual ERP power spectra of the cue-to-target interval when anticipating a visual target in the EEG-study.

Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 36 Hz (auditory frequency-tagging).

Suppl Fig. 10.

Exemplary illustration of the correlation between alpha power (0.5 – 0 s before target onset) and 36 Hz steady-state response (0.5 – 0 s before target onset) for each participant in block 2 (distractors present).

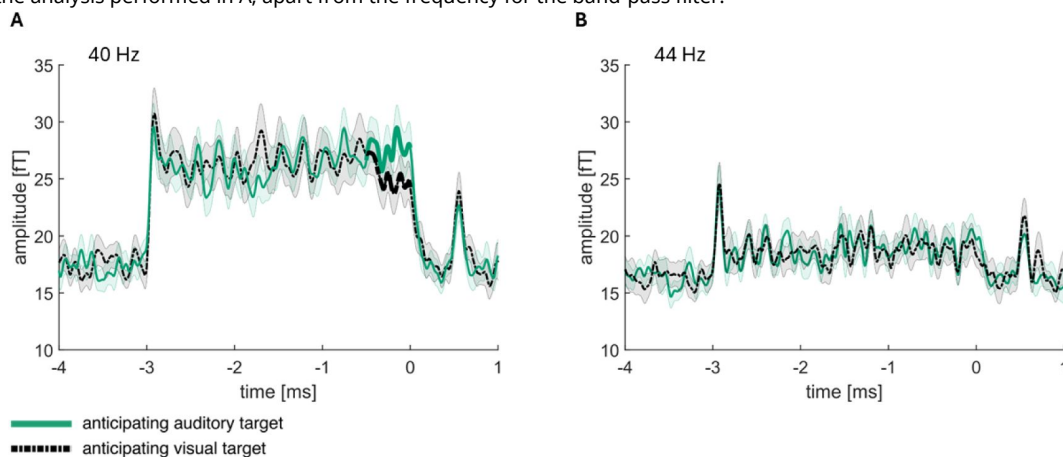
Alpha activity was averaged over the significant group difference cluster for alpha condition differences (seed cluster). Frequency-tagging activity was averaged over the significant cluster in the correlation with the alpha seed activity.

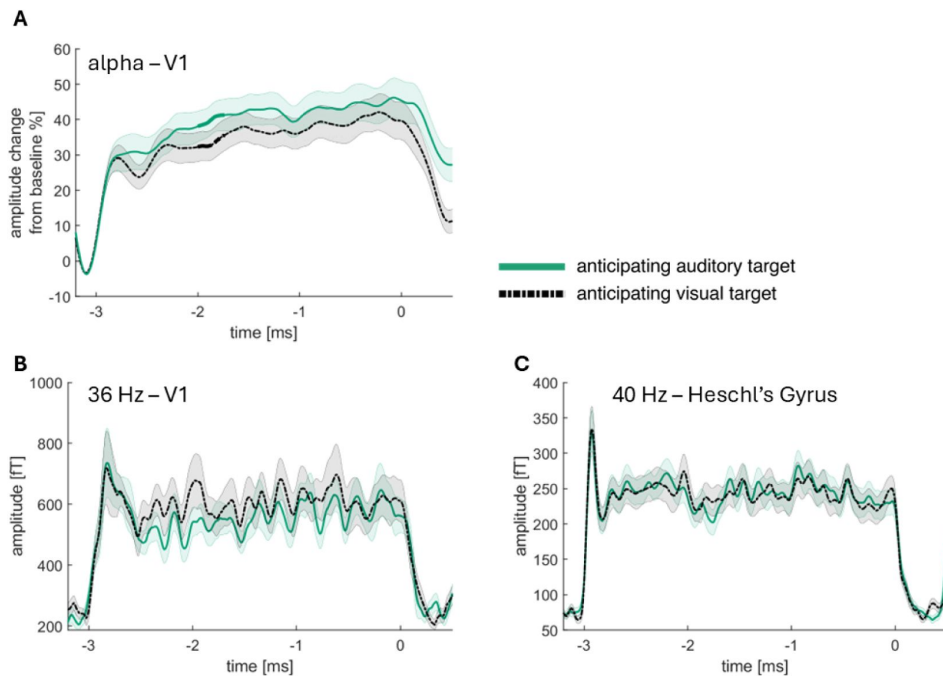


Suppl Fig. 11.

Illustration of bleeding over effects over a span of 4 Hz.

A, 40 Hz frequency-tagging data over the significant cluster differing between when expecting an auditory versus a visual target (identical to **Fig. 9** in the manuscript). B, 44 Hz signal over the same cluster chosen for A. The analysis was identical with the analysis performed in A, apart from the frequency for the band-pass filter.





Suppl Figure 12.

Virtual channels for V1 and Heschl's gyrus.

A, alpha power for the virtual channel created in V1 (Calcerine_L and Calcerine_R from AAL atlas; Tzourio-Mazoyer et al., 2002, NeuroImage). A cluster permutation analysis over time (between -2 and 0) revealed a significant condition difference between ~ -2 and -1.7 s ($p = 0.0449$). B, 36 Hz frequency-tagging signal for the virtual channel created in V1 (equivalent to the procedure in A). The same cluster permutation as performed in A revealed no significant condition differences. C, 40 Hz frequency-tagging signal for the virtual channel created in Heschl's gyrus (Heschl_L and Heschl_R from AAL atlas; Tzourio-Mazoyer et al., 2002, NeuroImage). The same cluster permutation as performed in A revealed no significant condition differences.

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Reviewer #1 (Public review):

In this study, Brickwedde et al. leveraged a cross-modal task where visual cues indicated whether upcoming targets required visual or auditory discrimination. Visual and auditory targets were paired with auditory and visual distractors, respectively. The authors found that during the cue-to-target interval, posterior alpha activity increased along with auditory and visual frequency-tagged activity when subjects were anticipating auditory targets. The authors conclude that their results disprove the alpha inhibition hypothesis, and instead implies that alpha "regulates downstream information transfer." However, as I detail below, I do not think the presented data irrefutably disproves the alpha inhibition hypothesis. Moreover, the evidence for the alternative hypothesis of alpha as an orchestrator for downstream signal transmission is weak. Their data serves to refute only the most extreme and physiologically implausible version of the alpha inhibition hypothesis, which assumes that alpha completely disengages the entire brain area, inhibiting all neuronal activity.

(1) Authors assign specific meanings to specific frequencies (8-12 Hz alpha, 4 Hz intermodulation frequency, 36 Hz visual tagging activity, 40 Hz auditory tagging activity), but the results show that spectral power increases in all of these frequencies towards the end of the cue-to-target interval. This result is consistent with a broadband increase, which could simply be due to additional attention required when anticipating auditory target (since behavioral performance was lower with auditory targets, we can say auditory discrimination was more difficult). To rule this out, authors will need to show a power spectral density curve with specific increases around each frequency band of interest. In addition, it would be more convincing if there was a bump in the alpha band, and distinct bumps for 4 vs 36 vs 40 Hz band.

(2) For visual target discrimination, behavioral performance with and without the distractor is not statistically different. Moreover, the reaction time is faster with distractor. Is there any evidence that the added auditory signal was actually distracting?

(3) It is possible that alpha does suppress task-irrelevant stimuli, but only when it is distracting. In other words, perhaps alpha only suppresses distractors that are presented simultaneously with the target. Since the authors did not test this, they cannot irrefutably reject the alpha inhibition hypothesis.

(4) In the abstract and Figure 1, the authors claim an alternative function for alpha oscillations; that alpha "orchestrates signal transmission to later stages of the processing stream." In support, the authors cite their result showing that increased alpha activity originating from early visual cortex is related to enhanced visual processing in higher visual areas and association areas. This does not constitute a strong support for the alternative hypothesis. The correlation between posterior alpha power and frequency-tagged activity was not specific in any way; Fig. 10 shows that the correlation appeared on both 1) anticipating-auditory and anticipating-visual trials, 2) the visual tagged frequency and the auditory tagged activity, and 3) was not specific to the visual processing stream. Thus, the data is more parsimonious with a correlation than a causal relationship between posterior alpha and visual processing.

<https://doi.org/10.7554/eLife.106050.2.sa3>

Reviewer #2 (Public review):

Brickwedde et al. investigate the role of alpha oscillations in allocating intermodal attention. A first EEG study is followed up with an MEG study that largely replicates the pattern of results (with small to be expected differences). They conclude that a brief increase in the amplitude of auditory and visual stimulus-driven continuous (steady-state) brain responses prior to the presentation of an auditory - but not visual - target speaks to the modulating role of alpha that leads them to revise a prevalent model of gating-by-inhibition.

Overall, this is an interesting study on a timely question, conducted with methods and analysis that are state-of-the-art. I am particularly impressed by the author's decision to replicate the earlier EEG experiment in MEG following the reviewer's comments on the original submission. Evidently, great care was taken to accommodate the reviewers suggestions.

In an earlier version, I was struggling with the report for two main reasons: It was difficult to follow the rationale of the study, due to structural issues with the narrative and missing information or justifications for design and analysis decisions, and I was not convinced that the evidence is strong, or even relevant enough for revising the mentioned alpha inhibition theory.

The authors have addressed my concerns through extensive revisions, and I find that it is now easier to follow, and makes a better case for rethinking how alpha may influence sensory processing through a clearer presentation of results and additional arguments.

<https://doi.org/10.7554/eLife.106050.2.sa2>

Reviewer #3 (Public review):

Brickwedde et al. attempt to clarify the role of alpha in sensory gain modulation by exploring the relationship between attention-related changes in alpha and attention-related changes in sensory-evoked responses, which surprisingly few studies have explicitly examined. The authors find evidence against the alpha-inhibition account, at least in early sensory processing, adding valuable data to the field to support our understanding of the alpha-inhibition hypothesis.

Due to task and measurement considerations, the EEG task is not sufficiently compelling to support the authors' claims that alpha inhibition does not occur in early sensory processing. However, the findings are bolstered by the additional MEG study which included changes in task design and a source-localization analysis. Importantly, the MEG results are aligned with the EEG study's key findings and support the authors' initial results, making a stronger case for their claims.

It is important to note that task designs can have great implications for the assessment of alpha inhibition, particularly with the use of stimuli that evoke a steady-state response, and the authors review these considerations during their discussion and interpretation of the theory. Overall, this paper is an excellent contribution to the alpha-inhibition literature and will hopefully motivate additional research on the specific relationship between these attention-related changes using both frequency-tagged and non-frequency-tagged stimuli in different task contexts.

<https://doi.org/10.7554/eLife.106050.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):

(1) Potential bleed-over across frequencies in the spectral domain is a major concern for all of the results in this paper. The fact that alpha power, 36Hz and 40Hz frequency-tagged amplitude and 4Hz intermodulation frequency power is generally correlated with one another amplifies this concern. The authors are attaching specific meaning to each of these frequencies, but perhaps there is simply a broadband increase in neural activity when anticipating an auditory target compared to a visual target?

We appreciate the reviewer's insightful comment regarding the potential bleed-over across frequencies in the spectral domain. We fully acknowledge that the trade-off between temporal and frequency resolution is a challenge, particularly given the proximity of the frequencies we are examining.

To address this concern, we performed additional analyses to investigate whether there is indeed a broadband increase in neural activity when anticipating an auditory target as compared to a visual target, as opposed to distinct frequency-specific effects. Our results show that the bleed-over between frequencies is minimal and does not significantly affect our findings. Specifically, we repeated the analyses using the same filter and processing steps for the 44 Hz frequency. At this frequency, we did not observe any significant differences between conditions.

These findings suggest that the effects we report are indeed specific to the 40 Hz frequency band and not due to a general broadband increase in neural activity. We hope this addresses the reviewer's concern and strengthens the validity of our frequency-specific results. We have now added this analysis to the methods section of our manuscript.

Line 730: To confirm that 4 Hz is a sufficient distance between tagging frequencies, we repeated to analysis for 43.5 to 44.5. We found no indication of frequency-bleeding over, as the effects observed at 40 Hz, were not present at 44 Hz (see SUPPL Fig. 11).

We do, however, not specifically argue against the possibility of a broadband increase in sensory processing when anticipating an auditory compared to a visual target. But even a broadband-increase would directly contradict the alpha inhibition hypothesis, which poses

that an increase in alpha completely disengage the whole cortex. We have made this clearer in the text now.

Line 491: As auditory targets were significantly more difficult than visual targets in our first study and of comparable difficulty in our second study, these results strongly speak to a vigilance increase of sensory processing independent of modality and an inability to selectively disengage one sensory modality in anticipation of a demanding task. This view is consistent with previous work in which visual SSEPs elicited by irrelevant background stimulation increased with task load in an auditory discrimination task (Jacoby et al., 2012).

(2) Moreover, 36Hz visual and 40Hz auditory signals are expected to be filtered in the neocortex. Applying standard filters and Hilbert transform to estimate sensory evoked potentials appears to rely on huge assumptions that are not fully substantiated in this paper. In Figure 4, 36Hz "visual" and 40Hz "auditory" signals seem largely indistinguishable from one another, suggesting that the analysis failed to fully demix these signals.

We appreciate the reviewer's insightful concern regarding the filtering and demixing of the 36 Hz visual and 40 Hz auditory signals, and we share the same reservations about the reliance on standard filters and the Hilbert transform method.

To address this, we would like to draw attention to SUPPL Fig. 11, which demonstrates that a 4 Hz difference is sufficient to effectively demix the signals using our chosen filtering and Hilbert transform approach. We argue that the reason the 36 Hz visual and 40 Hz auditory signals show similar topographies lies not in incomplete demixing but rather in the possibility that this condition difference reflects sensory integration, rather than signal contamination.

This interpretation is further supported by our findings with the intermodulation frequency at 4 Hz, which also suggests cross-modal integration. Furthermore, source localization analysis revealed that the strongest condition differences were observed in the precuneus, an area frequently associated with sensory integration processes. We have now expanded on this in the discussion section to better clarify this point.

Line 578: Previous research has shown that simultaneous frequency-tagging at multiple frequencies can evoke a response at the intermodulation frequency ($f_1 - f_2$), which in multimodal settings is thought to reflect cross-modal integration (Drijvers et al., 2021). This concept aligns closely with our findings, where increased vigilance in the sensory system, prompted by anticipation of a difficult auditory target, resulted in an increase in the intermodulation frequency. Similarly, our data shows that visual signal enhancement was localized in the precuneus, further supporting the role of this region in sensory integration (Al-Ramadhani et al., 2021; Xie et al., 2019).

(3) The asymmetric results in the visual and auditory modalities preclude a modality-general conclusion about the function of alpha. However, much of the language seems to generalize across sensory modalities (e.g., use of the term 'sensory' rather than 'visual').

We agree that in some cases we have not made a sufficient distinction between visual and sensory. We have now made sure, that when using 'sensory', we either describe overall theories, which are not visual-exclusive or refer to the possibility of a broad sensory increase. However, when directly discussing our results and the interpretation thereof, we now use 'visual'.

(4) In this vein, some of the conclusions would be far more convincing if there was at least a trend towards symmetry in source-localized analyses of MEG signals. For

example, how does alpha power in primary auditory cortex (A1) compare when anticipating auditory vs visual target? What do the frequency tagged visual and auditory responses look like when just looking at primary visual cortex (V1) or A1?

We thank the reviewer for this important suggestion and have added a virtual channel analysis. We were however, not interested in alpha power in primary auditory cortex, as we were specifically interested in the posterior alpha, which is usually increased when expecting an auditory compared to a visual target (and used to be interpreted as a blanket inhibition of the visual cortex). We have now improved upon the clarity concerning this point in the manuscript.

We have however, followed the reviewer's suggestion of a virtual channel analysis, showing that the condition differences are not observable in primary visual cortex for the 36 Hz visual signal and in primary auditory cortex for the 40 Hz auditory signal. Our data clearly shows that there is an alpha condition difference in V1, while there no condition difference for 36 Hz in V1 and for 40 Hz in Heschl's Gyrus.

Line 356: Additionally, we replicated this effect with a virtual channel analysis in V1 (see SUPPL Fig. 12)

Line 403: Furthermore, a virtual channel analysis in V1 and Heschl's gyrus confirmed that there were no condition differences in primary visual and auditory areas (see SUPPL Fig. 12).

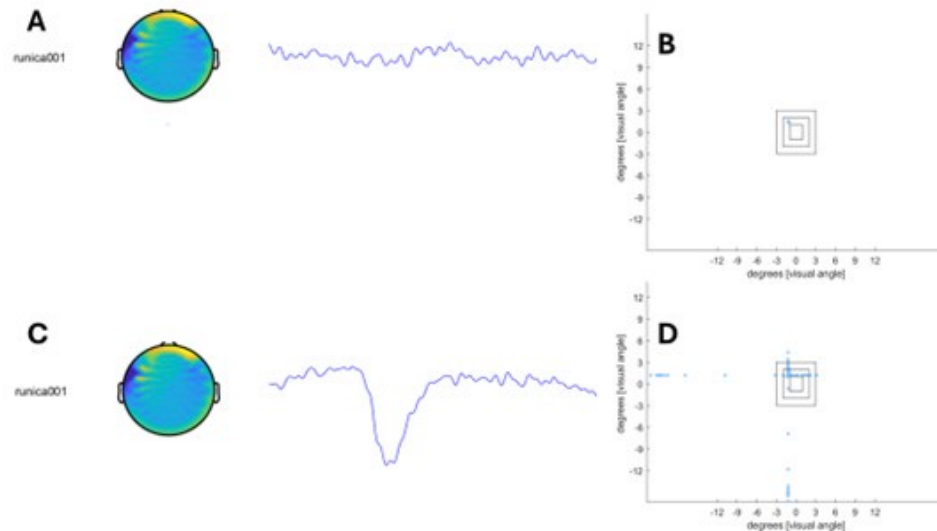
(5) Blinking would have a huge impact on the subject's ability to ignore the visual distractor. The best thing to do would be to exclude from analysis all trials where the subjects blinked during the cue-to-target interval. The authors mention that in the MEG experiment, "To remove blinks, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the data (See supplement Fig. 5)." This sentence needs to be clarified, since eye-movements cannot be measured during blinking. In addition, it seems possible to remove putative blink trials from EEG experiments as well, since blinks can be detected in the EEG signals.

We agree with the reviewer that this point has been phrased in a confusing way. From the MEG-data, we removed eyeblinks using ICA. Along for the supplementary Fig. 5 analysis, we used the eye-tracking data to make sure that participants were in fact fixating the centre of the screen. For this analysis, we removed trials with blinks (which can be seen in the eye-tracker as huge amplitude movements or as large eye-movements in degrees of visual angle; see figure below to show a blink in the MEG data and the according eye-tracker data in degrees of visual angle). We have now clarified this in the methods section.

As for the concern closed eyes to ignore visual distractors, in both experiments we can observe highly significant distractor cost in accuracy for visual distractors, which we hope will convince the reviewer that our visual distractors were working as intended.

Author response image 1.

Illustration of eye-tracker data for a trial without and a trial with a blink. All data points recorded during this trial are plottet. A, ICA component 1, which reflects blinks and its according data trace in a trial. No blink is visible. B, eye-tracker data transformed into degrees of visual angle for the trial depicted in A. C, ICA component 1, which reflects blinks and its according data trace in a trial. A clear blink is visible. D, eye-tracker data transformed into degrees of visual angle for the trial depicted in C.



Line 676: To confirm that participants had focused on the fixation cross during the cue-to-target interval, we incorporated eye-tracking into our MEG-experiment (EyeLink 1000 Plus). Correct trials of the second block were analysed for vertical and horizontal eye-movements. To exclude blinks from this analysis, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the eye-tracking data (See suppl Fig. 5).

(6) It would be interesting to examine the neutral cue trials in this task. For example, comparing auditory vs visual vs neutral cue conditions would be indicative of whether alpha was actively recruited or actively suppressed. In addition, comparing spectral activity during cue-to-target period on neutral-cue auditory correct vs incorrect trials should mimic the comparison of auditory-cue vs visual-cue trials. Likewise, neutral-cue visual correct vs incorrect trials should mimic the attention-related differences in visual-cue vs auditory-cue trials.

We have analysed the neutral cue trials in the EEG dataset (see suppl. Fig. 1). There were no significant differences to auditory or visual cues, but descriptively alpha power was higher for neutral cues compared to visual cues and lower for neutral cues compared to auditory cues. While this may suggest that for visual trials alpha is actively suppressed and for auditory trials actively recruited, we do not feel comfortable to make this claim, as the neutral condition may not reflect a completely neutral state. The neutral task can still be difficult, especially because of the uncertainty of the target modality.

As for the analysis of incorrect versus correct trials, we appreciate the idea, but unfortunately the accuracy rate was quite high so that the number of incorrect trials is insufficient to perform a reliable analysis.

(7) In the abstract, the authors state that "This implies that alpha modulation does not solely regulate 'gain control' in early sensory areas but rather orchestrates signal transmission to later stages of the processing stream." However, I don't see any supporting evidence for the latter claim, that alpha orchestrates signal transmission to later stages of the processing stream. If the authors are claiming an alternative function to alpha, this claim should be strongly substantiated.

We thank the reviewer for pointing out, that we have not sufficiently explained our case. The first point refers to gain control as elucidated by the alpha inhibition hypothesis, which claims that increases in alpha disengage an entire cortical area. Since we have confirmed the alpha increase in our data to originate from primary visual cortex through source analysis, this should lead to decreased visual processing. The increase in 36 Hz visual processing therefore directly contradicts the alpha inhibition hypothesis. We propose an alternative explanation for the functionality of alpha activity in this task. Through pulsed inhibition, information packages of relevant visual information could be transmitted down the processing stream, thereby enhancing relevant visual signal transmission. We argue the fact that the enhanced visual 36 Hz signal we found correlated with visual alpha power on a trial-by-trial basis, and did not originate from primary visual cortex, but from areas known for sensory integration supports our claim.

We have now tried to make this point clearer by rephrasing our manuscript. Additionally, we have also now further clarified this point in our discussion.

Line 527: Our data provides evidence in favour of this view, as we can show that early sensory alpha activity covaries over trials with SSEP magnitude in higher order sensory areas. If alpha activity exerted gain control in early visual regions, increased alpha activity would have to lead to a decrease in SSEP responses. In contrast, we observe that increased alpha activity originating from early visual cortex is related to enhanced visual processing. Source localization confirmed that this enhancement was not originating from early visual areas, but from areas associated with later stages of the processing stream such as the precuneus, which has been connected to sensory integration (Al-Ramadhani et al., 2021; Xie et al., 2019). While we cannot completely rule out alternative explanations, it seems plausible to assume that inhibition of other task-irrelevant communication pathways leads to prioritised and thereby enhanced processing over relevant pathways. In line with previous literature (Morrow et al., 2023; Peylo et al., 2021; Zhigalov & Jensen, 2020b), we therefore suggest that alpha activity limits task-irrelevant feedforward communication, thereby enhancing processing capabilities in relevant downstream areas (see Fig. 1A).

Reviewer #1 (Recommendations for the authors): Minor Concerns:

- (1) I suggest adding more details about the task in the Results and/or Figure 1 legend. Specifically, when describing the task, I think it would help the readers if the authors specified what the participants had to do to get a trial correct (e.g., press left / down / right arrow if the tone pitch was low (500Hz) / medium (1000Hz) / high (2000Hz).)
- (2) Please clarify whether Gaboar patch was drifting.
- (3) Figure 2C-D: I suggest clarifying in the X-tick labels that + and - trials are in separate blocks (e.g., put 'Block1 visual-' instead of 'visual-').

We followed the suggestions of the reviewer detailed in point 1-3, which indeed greatly improves the clarity and readability of these parts.

- (4) "Interestingly, auditory distractors reduced reaction times to visual targets, which could be explained by a generally faster processing of auditory targets (Jain et al., 2015), possibly probing faster responses in visual tasks (Naue et al., 2011)." - Please elaborate on how faster processing of auditory targets could lead to the probing of faster responses in visual tasks. Further, if I understand correctly, this should result in a speed-accuracy trade-off, which is not observed in the MEG experiments. If there is a learning effect due to the blocked structure in the MEG experiments, why is it not observed on auditory trials?

We thank the reviewer for suggesting clarifying this paragraph. We have now rephrased this part and added additional information.

Concerning the reviewer's theory, intersensory facilitation can occur in the absence of a speed-accuracy trade-off, as it can affect the motor execution after a decision has been made. Nevertheless, learning effects could also have led to this result in the MEG experiment. Our difficulty calibration did not lead to comparable accuracies in block 1, where auditory targets were now less difficult than visual targets. With the addition of distractors in block 2, accuracy for auditory targets decreased, while it increased for visual targets. Indeed, one interpretation could be that there was a learning effect for visual targets, which was not prevalent for auditory targets. However, the speed increase when visual targets are coupled with auditory distractors is prevalent in both experiments. Accordingly, we find the intersensory facilitation account more likely.

line 148: Interestingly, auditory distractors reduced reaction times to visual targets, which could be explained by a generally faster processing of auditory targets (Jain et al., 2015). As such, the auditory distractor possibly caused intersensory facilitation (Nickerson., 1973), whereby reaction times to a target can be facilitated when accompanied by stimuli of other sensory modalities, even if they are irrelevant or distracting.

(5) Please briefly describe the cluster permutation analysis in the results section.

We have now added a brief description of the cluster permutation analysis we performed in the results section.

Line 166: We then applied cluster permutation analysis, whereby real condition differences were tested against coincidental findings by randomly permutating the condition labels to the data and testing for condition differences 1000 times (Maris & Oostenveld, 2007).

(6) Figure 4A legend: "auditory steady-state evoked potential (ASSEP) averaged over 6 central electrodes displaying the highest 40 Hz power (Fz, FC1, FC2, F11, F2, FCz)." - I suggest marking these 6 electrodes in the scalp map on the figure panel.

We have followed the suggestion of the reviewer and marked the electrodes/sensors used to illustrate the steady-state responses.

(7) Lines 281-283: "It was highly significant for the visual 36 Hz response (Fig. 5A, middle columns, $p = .033$; $t(19) = 2.29$; $BF(10) = 1.91$) but did not reach significance for the visual 40 Hz response (Fig. 5B, middle column; $p = 0.20$; $t(19) = 1.32$; $BF(10) = 0.49$)."- Was "visual 40Hz response" a typo? I believe 40Hz pertains to auditory, not visual?

We thank the reviewer for pointing out this error and agree that the phrasing was sometimes confusing. We have now used the terms VSSEP and ASSEP to make things clearer throughout the manuscript.

L. 224-229: The median split was highly significant for the 36 Hz VSSEP response (Fig. 5A, middle columns, $p = .033$; $t(19) = 2.29$; $BF(10) = 1.91$) but did not reach significance for the 40 Hz ASSEP response (Fig. 5B, middle column; $p = 0.20$; $t(19) = 1.32$; $BF(10) = 0.49$).

Reviewer #2 (Public review):

Brickwedde et al. investigate the role of alpha oscillations in allocating intermodal attention. A first EEG study is followed up with an MEG study that largely replicates the pattern of results (with small to be expected differences). They conclude that a brief increase in the amplitude of auditory and visual stimulus-driven continuous (steady-

state) brain responses prior to the presentation of an auditory - but not visual - target speaks to the modulating role of alpha that leads them to revise a prevalent model of gating-by-inhibition.

Overall, this is an interesting study on a timely question, conducted with methods and analysis that are state-of-the-art. I am particularly impressed by the author's decision to replicate the earlier EEG experiment in MEG following the reviewer's comments on the original submission. Evidently, great care was taken to accommodate the reviewers suggestions.

We thank the reviewer for the positive feedback and expression of interest in the topic of our manuscript.

Nevertheless, I am struggling with the report for two main reasons: It is difficult to follow the rationale of the study, due to structural issues with the narrative and missing information or justifications for design and analysis decisions, and I am not convinced that the evidence is strong, or even relevant enough for revising the mentioned alpha inhibition theory. Both points are detailed further below.

We have now revised major parts of the introduction and results in line with the reviewer's suggestions, hoping that our rationale is now easier to follow and that our evidence will now be more convincing. We have separated our results section into the first study (EEG) and to second study (MEG), to enhance the rationale of our design choices and readability. We have clarified all mentioned ambiguous parts in our methods section. Additionally, we have revised the introduction to now explain more clearly what results to expect under the alpha inhibition theory in contrast to our alternative account.

Strength/relevance of evidence for model revision: The main argument rests on 1) a rather sustained alpha effect following the modality cue, 2) a rather transient effect on steady-state responses just before the expected presentation of a stimulus, and 3) a correlation between those two. Wouldn't the authors expect a sustained effect on sensory processing, as measured by steady-state amplitude irrespective of which of the scenarios described in Figure 1A (original vs revised alpha inhibition theory) applies? Also, doesn't this speak to the role of expectation effects due to consistent stimulus timing? An alternative explanation for the results may look like this: Modality-general increased steady-state responses prior to the expected audio stimulus onset are due to increased attention/vigilance. This effect may be exclusive (or more pronounced) in the attend-audio condition due to higher precision in temporal processing in the auditory sense or, vice versa, too smeared in time due to the inferior temporal resolution of visual processing for the attend-vision condition to be picked up consistently. As expectation effects will build up over the course of the experiment, i.e., while the participant is learning about the consistent stimulus timing, the correlation with alpha power may then be explained by a similar but potentially unrelated increase in alpha power over time.

We thank the reviewer for raising these insightful questions and suggestions.

It is true that our argument rests on a rather sustained alpha effect and a rather transient effect on steady-state responses, and a correlation between the two. However, this connection would not be expected under the alpha inhibition hypothesis, which states that alpha activity would inhibit a whole cortical area (when irrelevant to the task), exerting “gain control”. This notion directly contradicts our results of the “irrelevant” visual information a) being transmitted at all and b) increasing.

However, it has been shown in various reports (see for instance Dugué et al., 2011; Haegens et al., 2011; Spaak et al., 2012) that alpha activity exerts pulsed inhibition, so we proposed an alternative theory of an involvement in signal transmission. In this case, the cyclic inhibition would serve as an ordering system, which only allows for high-priority information to pass, resulting in higher signal-to-noise ratio. We do not make a claim about how fast or when these signals are transmitted in relation to alpha power. For instance, it could be that alpha power increases as a preparatory state even before signal is actually transmitted. Zhigalov (2020 Hum. Brain M.) has shown that in V1, frequency-tagging responses were up-and down regulated with attention – independent of alpha activity.

However, we do believe that visual alpha power correlates on a trial-by-trial level with visual 36 Hz frequency-tagging increases (see Fig. 5 and 10 in our manuscript) - a relationship which has not been found in V1 by us and others (see SUPPL Fig. 12 and Zhigalov 2020, Hum. Brain Mapp.) suggest a strong connection. Furthermore, the fact that the alpha modulation originates from early visual areas and occurs prior to any frequency-tagging changes, while the increase in frequency-tagging can be observed in areas which are later in the processing stream (such as the precuneus) is strongly indicative for an involvement of alpha power in the transmission of this signal. We cannot fully exclude alternative accounts and mechanisms which effect both alpha power and frequency-tagging responses.

The alternative account described by the reviewer does not contradict our theory, as we argue that the alpha power modulation reflects an expectation effect (and the idea that it could be related to the resolution of auditory versus visual processing is very interesting!). It is also possible that this expectation is, as the reviewer suggests, related to attention/vigilance and might result in a modality-general signal increase. By way of support, we observed an increase in the frequency-tagging response in sensory integration areas. Accordingly, we argue that the alternative explanation provided by the reviewer contradicts the alpha inhibition hypothesis, but not necessarily our alternative theory.

We have now revised the discussion and are confident our case is now stronger and easier to follow. Additionally, we mentioned the possibility for alternative explanations as well as the possibility, that alpha networks fulfil different roles in different locations/task environments.

Line 523: Here we propose that alpha activity, rather than modulating early primary sensory processing, exhibits its inhibitory effects at later stages of the processing stream (Antonov et al., 2020; Gundlach et al., 2020; Zhigalov & Jensen, 2020a; Zumer et al., 2014), gating feedforward or feedback communication between sensory areas (Bauer et al., 2020; Haegens et al., 2015; Uemura et al., 2021). Our data provides evidence in favour of this view, as we can show that early sensory alpha activity covaries over trials with SSEP magnitude in higher order sensory areas. If alpha activity exerted gain control in early visual regions, increased alpha activity would have to lead to a decrease in SSEP responses. In contrast, we observe that increased alpha activity originating from early visual cortex is related to enhanced visual processing. Source localization confirmed that this enhancement was not originating from early visual areas, but from areas associated with later stages of the processing stream such as the precuneus, which has been connected to sensory integration (Al-Ramadhani et al., 2021; Xie et al., 2019). While we cannot completely rule out alternative explanations, it seems plausible to assume that inhibition of other task-irrelevant communication pathways leads to prioritised and thereby enhanced processing over relevant pathways. In line with previous literature (Morrow et al., 2023; Peylo et al., 2021; Zhigalov & Jensen, 2020b), we therefore suggest that alpha activity limits task-irrelevant feedforward communication, thereby enhancing processing capabilities in relevant downstream areas (see Fig. 1A).

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Zhigalov, A., & Jensen, O. (2020). Alpha oscillations do not implement gain control in early visual cortex but rather gating in parieto-occipital regions. *Human Brain Mapping*, 41(18), 5176–5186. <https://doi.org/10.1002/hbm.25183>

Structural issues with the narrative and missing information: Here, I am mostly concerned with how this makes the research difficult to access for the reader. I list the some major, followed by more specific points below:

In the introduction the authors pit the original idea about alpha's role in gating against some recent contradictory results. If it's the aim of the study to provide evidence for either/or, predictions for the results from each perspective are missing. Also, it remains unclear how this relates to the distinction between original vs revised alpha inhibition theory (Fig. 1A). Relatedly, if this revision is an outcome rather than a postulation for this study, it shouldn't be featured in the first figure.

We agree with the reviewer that we have not sufficiently clarified our goal as well as how different functionalities of alpha oscillations would lead to different outcomes. We have revised the introduction and restructured the results part and hope that it is now easier to follow. The results part now follows study 1 (EEG) and study 2 (MEG) chronologically, so that results can more easily be differentiated and our design choices for the second study can be explained better.

Line 50: Recent evidence challenged a direct connection between alpha activity and visual information processing in early visual cortex. As such, both visual steady-state responses and alpha power were modulated by attention, but did not covary when investigating individual trials (Zhigalov & Jensen, 2020). Unfortunately, very few studies have investigated direct connections between alpha activity, attention and sensory signals, especially over trials. Furthermore, results seem to depend on timing of alpha activity in relation to sensory responses as well as stimulus type and outcome measure (Morrow et al., 2023).

Accordingly, the objective of the current study is to test the alpha inhibition hypothesis compared to an alternative theory. Based on the alpha inhibition hypothesis, alpha modulation is connected to ‘gain control’ in early visual areas through modulation of excitability (Foxy & Snyder, 2011; Jensen & Mazaheri, 2010; Van Diepen et al., 2019). In contrast, we propose that inhibitory effects of alpha modulation are exhibited at later stages of the processing stream (Peylo et al., 2021; Yang et al., 2023; Zhigalov & Jensen, 2020a; Zumer et al., 2014), gating feedforward or feedback communication between sensory areas (see Fig. 1B; Bauer et al., 2020; Haegens et al., 2015; Uemura et al., 2021).

Line 80: The aim of our study was to directly test the alpha inhibition hypothesis by investigating if cue-induced modulation of alpha activity coincides with the suppression of frequency-tagging responses in task-irrelevant modalities.

Line 99: In brief, while we observed the expected cue-induced early-visual alpha modulation, the amplitude of auditory and visual SSEP/SSEFs as well as their intermodulation frequency increased just prior to the onset of the auditory target, contradicting the alpha inhibition hypothesis. The difference between conditions of visual SSEP/SSEFs originated from sensory integration areas and correlated with early sensory alpha activity on a trial-by-trial basis, speaking to an effect of alpha modulation on signal transmission rather than inhibition of early visual areas.

The analysis of the intermodulation frequency makes a surprise entrance at the end of the Results section without an introduction as to its relevance for the study. This is provided only in the discussion, but with reference to multisensory integration, whereas the main focus of the study is focussed attention on one sense. (Relatedly, the reference to "theta oscillations" in this sections seems unclear without a reference to the overlapping frequency range, and potentially more explanation.) Overall, if there's no immediate relevance to this analysis, I would suggest removing it.

We thank the reviewer for pointing this out and have now added information about this frequency to the introduction. We believe that the intermodulation frequency analysis is important, as it potentially supports the notion that condition differences in the visual-frequency tagging response are related to downstream processing rather than overall visual information processing in V1. We would therefore prefer to leave this analysis in the manuscript.

Line 75: Furthermore, when applying two different frequencies for two different sensory modalities, their intermodulation frequency (f_1 - f_2) has been suggested to reflect cross-modal integration (Drijvers et al., 2021). Due to distinct responses, localisation and attention-dependence, frequency-tagging provides an optimal tool to study sensory signal processing and integration over time.

Reviewer #2 (Recommendations for the authors):

As detailed in several points below, I found that I didn't get the information I needed to fully understand design/analysis decisions. In some cases, this may just be a case of re-organising the manuscript, in others crucial info should be added:

Specific issues:

Page 2, line 51: How does recent evidence contradict this? Please explain.

We have added a section that describes the results contradicting the alpha inhibition hypothesis.

Line 50: Recent evidence challenged a direct connection between alpha activity and visual information processing in early visual cortex. As such, both visual steady-state responses and alpha power were modulated by attention, but did not covary when investigating individual trials (Zhigalov & Jensen, 2020).

Page 3, line 78-80: "... also interested in relationships [...] on a trial-by-trial basis" - why? Please motivate.

We thank the reviewer for highlighting this section, which we feel was not very well phrased. We have rewritten this whole paragraph and hope that our motivation for this study is now clear.

Line 50: Recent evidence challenged a direct connection between alpha activity and visual information processing in early visual cortex. As such, both visual steady-state responses and alpha power were modulated by attention, but did not covary when investigating individual trials (Zhigalov & Jensen, 2020). Unfortunately, very few studies have investigated direct connections between alpha activity, attention and sensory signals, especially over trials. Furthermore, results seem to depend on timing of alpha activity in relation to sensory responses as well as stimulus type and outcome measure (Morrow et al., 2023).

Page 4, line 88-92: "... implementing a blocked design" - unclear why? This is explained to some extent in the next few lines but remains unclear without knowing outcomes of the EEG experiment with more detail. Overall, it seems like this methodological detail may be better suited for a narrative in the Results section, that follows a more chronological order from the findings of the EEG experiment to the design of the MEG study.

More generally, and maybe I missed it, I couldn't find a full account of why a block design was chosen and what the added value was. I believe that re-organising the Results section would allow precisely stating how that was an improvement over the EEG experiment.

In line with the reviewer's suggestion, we have now restructured the results section. The first section of the study 2 results now explains our design choices with direct reference to the results of the EEG experiment.

Line 298: To test the robustness of our results and to employ additional control analyses, we replicated our experiment using MEG (see Fig. 7A). While an increase in visual information processing parallel to an increase in alpha modulation already contradicts the notion of alpha inhibition exerting "gain control", affecting the whole visual cortex, our claim that alpha modulation instead affects visual information at later processing stages still required further validation. As such, our goal was to perform source analyses showing alpha modulation originating from primary visual areas affected visual information at later processing stages (e.g. not in primary visual cortex). Additionally, to exclude that the uncertainty over possible distractors affected our results, we employed a block design, where block 1 consisted only of trials without distractors and in block 2 targets were always accompanied by a distractor. Furthermore, we aligned the visual and auditory task to be more similar, both of them now featuring frequency-discrimination, which related to sound pitch (frequency) in the auditory condition and stripe-frequency of the Gabor patch in the visual condition. Lastly, to make sure our effects were driven by sensory modality-differences rather than task-difficulty differences, we included a short calibration phase. Prior to the experiment, difficulty of pitch sounds, and Gabor patch frequency were calibrated for each individual, ascertaining a success rate between 55% to 75%.

The point above also applies to lines 95-97 where it's unclear what "aligning the visual with the auditory task" means. Also, what would be the predictions for "more nuanced interactions [...]"

We agree that this phrasing was more than confusing and in the process of restructuring our results section, we have now revised this passage (see cited text from our manuscript to the point just above).

Page 9, line 207-209: One of the few mentions of the "ambivalent" condition (attention to audio+vision?). To what end was that condition added to the experiment originally? The explanation that this condition was dropped from analysis because it did not show significant results does not seem methodologically sound.

We thank the reviewer for pointing this out, as we had changed the name from ambivalent to non-specific, but this word had slipped our attention. The condition was added to the experiment as a control, which enables us to verify that our cues as well as our distractors work as intended. While interesting to analyse (and we did not drop it completely, the condition comparisons are in the supplementary material), we felt that further analysis of this condition would not contribute to addressing our research question. To be specific, the prerequisite to analysing the effect of alpha modulation is a significant effect of alpha modulation in the first place. We have now clarified the rationale for this condition, as well as our reasoning for omitting it from correlation and source analysis.

Line 173 When presenting unspecified cues, alpha power changes were not significant, but descriptively larger compared to visual target conditions and lower compared to auditory target conditions (see suppl Fig. 2). However as significant alpha modulation was a prerequisite to test our hypotheses, we excluded this condition from further analysis.

Page 9, line 209-212: "condition differences in alpha were only significant in block 2 [...] therefore we performed the [...] analysis [...] only for the second half of the experiment." This sounds like double-dipping. Maybe just an issue of phrasing?

We thank the reviewer for pointing out that it may appear like ‘double dipping’. The reasoning was the same as the point above, we require a significant alpha modulation to test the effect of alpha modulation on further processing. We have revised this part to be clearer.

Line 345: In line with previous studies (van Diepen & Mazaheri, 2017), condition differences in alpha activity were only significant in block 2, where distractors were present. As alpha modulation was a prerequisite to test our hypotheses, we performed the following analyses solely with data from block 2 (see Fig. 8).

Page 12, line 281: Bayes factors are used here (and elsewhere), in addition to NHST. May be worthwhile to mention that briefly before use and give an intro sentence on its use, value and interpretation, and why these are added sometimes but not for all tests reported.

We agree that we did not introduce this at all and have now added a section, which explains the inclusion as well as the interpretation of the Bayes factor.

Line 218: To estimate the robustness of these results, we additionally conducted median split analyses between trials with high and low alpha power for each participant, as well as averaged the correlation coefficient of each participant and calculated a one-sample t-test against 0. For each analysis we provided the Bayes Factor, which estimates the strength of support for or against the null hypothesis (BF > 3.2 is considered as substantial evidence and BF > 10 is considered as strong evidence; Kass & Raftery, 1995).

Throughout the Results section, it's not always clear which results are from the EEG or from the MEG study. Adopting the recommendation in point c) may help with that.

According to the reviewer’s recommendation, we have restructured our results section and first present the EEG study and afterwards the MEG study.

Similarly, it seems pivotal to add "visual" and "auditory" when mentioning the 36/40-Hz steady-state responses (or stimulation) to help the reader.

We agree that visual/auditory 36 Hz / 40 Hz frequency-tagging responses, expecting visual/auditory target becomes lengthy and confusing very quickly. We therefore decided to

introduce the abbreviation of visual steady-state evoked potentials/fields (VSSEP/VSEF) and auditory steady-state evoked potentials/fields (ASSEP/ASSEF).

Figure 5 - showing the same cluster as "early" and "late" in the margin for the MEG data is potentially confusing.

We thank the reviewer for pointing this out and have now adapted the figure to just show one cluster, as we only found this one cluster in our MEG analysis.

Reviewer #3 (Public review):

This paper seems very strong, particularly given that the follow-up MEG study both (a) clarifies the task design and separates the effect of distractor stimuli into other experimental blocks, and (b) provides source-localization data to more concretely address whether alpha inhibition is occurring at or after the level of sensory processing, and (c) replicates most of the EEG study's key findings.

We thank the reviewer for their positive feedback and evaluation of our work.

There are some points that would be helpful to address to bolster the paper. First, the introduction would benefit from a somewhat deeper review of the literature, not just reviewing when the effects of alpha seem to occur, but also addressing how the effect can change depending on task and stimulus design (see review by Morrow, Elias & Samaha (2023)).

We thank the reviewer for this suggestion and agree. We have now added a paragraph to the introduction that refers to missing correlation studies and the impact of task design.

Line 53: Unfortunately, very few studies have investigated direct connections between alpha activity, attention and sensory signals, especially over trials. Furthermore, results seem to depend on timing of alpha activity in relation to sensory responses as well as stimulus type and outcome measure (Morrow et al., 2023).

Additionally, the discussion could benefit from more cautionary language around the revision of the alpha inhibition account. For example, it would be helpful to address some of the possible discrepancies between alpha and SSEP measures in terms of temporal specificity, SNR, etc. (see Peylo, Hilla, & Sauseng, 2021). The authors do a good job speculating as to why they found differing results from previous cross-modal attention studies, but I'm also curious whether the authors think that alpha inhibition/modulation of sensory signals would have been different had the distractors been within the same modality or whether the cues indicated target location, rather than just modality, as has been the case in so much prior work?

We thank the reviewer for suggesting these interesting discussion points and have included a paragraph in our discussion that clarifies these issues.

Line 543: It should be noted, the comparison between modulation in alpha activity and in SSEP/SSEFs is difficult, especially concerning timing. This is largely owed to differences in signal-to-noise due to trial averaging in the frequency versus the time domain and temporal and frequency lag in the estimation of alpha activity (Peylo et al., 2021). It is further noteworthy, that the majority of evidence for the alpha inhibition hypothesis focused on the effect of pre-target alpha modulation on behaviour and target-related potentials (Morrow et al., 2023). However, in our data alpha modulation occurs clearly ahead of SSVEP/SSVEF modulation on a scale that could not be simply explained by temporal or frequency smearing.

Additionally, significant trial-by-trial correlations, which occur in the frequency domain for both signal types, underline the strong relationship between both measurements.

Interestingly, we could show that the magnitude of the correlation between alpha power and visual information processing varied between conditions, suggesting a dynamic and adaptive regime. This notion supports the view that alpha oscillations represent a mechanism rather than a specific function, which can fulfil different roles depending on task demand and network location, which has been confirmed in a recent study revealing functionally distinct alpha networks (Clausner et al., 2024). As such, it is conceivable that alpha oscillations can in some cases inhibit local processing, while in other cases, depending on network location, connectivity and demand, alpha oscillation can facilitate signal transmission. In different contexts, utilizing unimodal targets and distractors, spatial cueing, or covert attention, different functional processes could be involved (Morrow et al., 2023). Future research should intensify efforts to disentangle these effects, investigating localized alpha networks intracranially or through combinations of fMRI, EEG and MEG, to clearly measure their effects on sensory processing and behaviour.

Overall, the analyses and discussion are quite comprehensive, and I believe this paper to be an excellent contribution to the alpha-inhibition literature.

Reviewer #3 (Recommendations for the authors):

Overall, the paper is well-written, and the analyses and interpretations are strong. I think that the end of the introduction would feel more complete and more read more easily if you outlined all of your main hypotheses (not just trials signaling an auditory stimulus, but visual trials too, and what about distractor trials? This could help justify changes to task design in the MEG study), and then the key findings that motivated the follow-up design, which you then discuss (as opposed to introducing a new aim in this paragraph).

We thank the reviewer for this positive evaluation. Based on feedback und suggestions from all reviewers, we have revised the structure of the manuscript. The introduction now states more clearly which results would be expected under the alpha inhibition theory and how our results contradict this. The results section has now been divided into two studies, which will make the rationale for our follow-up design easier to follow.

Line 80: The aim of our study was to directly test the alpha inhibition hypothesis by investigating if cue-induced modulation of alpha activity coincides with the suppression of frequency-tagging responses in task-irrelevant modalities.

Line 96: In brief, while we observed the expected cue-induced early-visual alpha modulation, the amplitude of auditory and visual SSEP/SSEFs as well as their intermodulation frequency increased just prior to the onset of the auditory target, contradicting the alpha inhibition hypothesis. The difference between conditions of visual SSEP/SSEFs originated from sensory integration areas and correlated with early sensory alpha activity on a trial-by-trial basis, speaking to an effect of alpha modulation on signal transmission rather than inhibition of early visual areas.

Minor issues:

L84 - "is" should be "was"

L93 - "allows" should be "allowed"

L113 - I think "changed" would suffice

Fig 1A (text within figure on top) - "erea" should be "area" and caption title should include "of" (Illustration of the...)

L213 - time window could be clarified

Fig 4 -captions inconsistently capitalize words and use) and , following the caption letters

L253-255 - give you are looking at condition differences, do you mean the response was larger before an auditory target than before a visual target? It currently reads as if you mean that it was larger in that window right before the target as opposed to other time windows

L368 - "behaviorally" should be "behavioral"

L407-408 - I think auditory SSEP/SSVEFs should be auditory or visual SSEP/SSEFs, unless you are specifically only talking about auditory SSEPs and visual SSEFs

L411 - also uses SSVEFs

L413 - "frequently, or in the case of..."

L555 - "predicting" should be predicted? Or do you mean only cues that correctly predicted the target?

We are very grateful for the reviewer for pointing out these mistakes, all of which we have remedied in our manuscript.

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